

Honeywell

TPS System and Process Operations Guide

TP07X-420

R420

9/2009

Notices and Trademarks

**Copyright 2009 by Honeywell International Inc.
Release R420 September 2009**

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell **TotalPlant** and Experion are registered trademarks of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell International
Process Solutions
2500 West Union Hills
Phoenix, AZ 85027
1-800 343-0228

About This Document

This document describes the following:

- TPS System Operations, including the following Native Window-based operations:
 - Logging onto a TPS node
 - Using the Signon Manager
 - Loading the LCN/TPN personality
 - Shutting down the LCN/TPN personality
- TPS Process Operations, including the following non-Native Window operations:
 - Generating LCN Revision and Version Reports
 - Using the GUS Logic tool

Release Information

Document Name	Document ID	Release Number	Publication Date
TPS System and Process Operations Guide - TP07X	TP07X-420	420	9/2009

Document Category
Operations

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Document ID
SafeView User's Guide	GU27
Native Window User's Guide	GU20
TPS System Administration Guide	TP06X
TPS System Implementation Guide	TP08X
LCN/TPN Process Operations Manual	SW11-601
System Management Operations Guide	EX07
Experion Operators Guide	EP-DSXX44
Experion Server and Client Configuration Guide	EP-DSXX24

Support and Other Contacts

Location	Organization	Address	Phone/Facsimile
United States and Canada	Honeywell Solution Support Center	Honeywell TAC, MS P13, 2500 West Union Hills Drive, Phoenix, AZ, 85027	Phone: 1-800 822-7673 602- 313-5558 (In Arizona) Facsimile: (602) 313-3293
Note: Calls are answered by dispatcher between 6:00 A.M. and 4:00 P.M. Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.			
Europe	Honeywell TAC-EMEA	TAC-BE02 Hermes Plaza Hermeslaan, 1HB-1831 Diegem, Belgium	Phone: +32-2-728-2732 Facsimile: +32-2-728-2696
Pacific	Honeywell Global TAC – Pacific	Honeywell Limited Australia 5 Kitchener Way Burswood 6100, Western Australia Email: GTAC@honeywell.com	Phone: 1300-300-4822 (toll free within Australia) +61-8-9362-9559 (outside Australia) Facsimile: +61-8-9362-9564
India	Honeywell Global TAC – India	Honeywell Automation India Ltd. 56 and 57, Hadapsar Industrial Estate Hadapsar, Pune –411 013, India Email: Global-TAC- India@honeywell.com	Phone: +91-20- 66039400 Facsimile: +91-20- 66039800

Support and Other Contacts

Location	Organization	Address	Phone/Facsimile
Korea	Honeywell Global TAC – Korea	Honeywell Co., Ltd 4F, Sangam IT Tower B4-4 Block 1590, DMC Sangam-dong, Mapo-gu, Seoul, 121-836, Korea Email: Global-TAC- Korea@honeywell.com	Phone: +82-2-799-6317 Facsimile: +82-2-792-9015
People's Republic of China	Honeywell Global TAC – China	Honeywell (China) Co., Ltd 33/F, Tower A, City Center, 100 Zunyi Rd., Shanghai 200051, People's Republic of China Email: Global-TAC- China@honeywell.com	Phone: +86- 21-52574568
Singapore	Global TAC – South East Asia	Honeywell Private Limited Honeywell Building 17, Changi Business Park Central 1, Singapore 486073 Email: GTAC- SEA@honeywell.com	Phone: +65-6580-3500 Facsimile: +65-6580-3501 +65-6445-3033
Taiwan	Global TAC – Taiwan	Honeywell Taiwan Ltd. 17F-1, No. 260, Jhongshan 2nd Road. Cianjhen District Kaohsiung, Taiwan, ROC Email: Global-TAC- Taiwan@honeywell.com	Phone: +886- 7- 536 2567 Facsimile: +886-7-536 2039

Location	Organization	Address	Phone/Facsimile
Japan	Global TAC – Japan	Honeywell Japan Inc. New Pier Takeshiba, South Tower Building, 20th Floor, 1-16-1 Kaigan, Minato-ku, Tokyo 105-0022, Japan Email: Global-TAC- JapanJA25@honeywell. com	Phone: +81-3-6730-7160 Facsimile: +81-3-6730-7228
Elsewhere	Call your nearest Honeywell office.		
World Wide Web	Honeywell Solution Support Online: http://www.honeywell.com/ps		
Training Classes	Honeywell Automation College: http://www.automationcollege.com		

Symbol Definitions

The following table lists those symbols used in this cument to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

Contents

1. INTRODUCTION	13
1.1 Node fundamentals	13
LCNP board	13
Native Window	13
Node startup and log on	14
Access level and Signon Manager	14
TPS managed components	14
TPS node startup flowchart	16
1.2 Status displays	18
System Status display	18
System Management display	19
2. LOGGING ON TO A TPS NODE	21
2.1 Logon Requirements for TPS nodes	21
Custom TPS user accounts	21
2.2 TPS node log on procedures	22
Logon procedures	22
3. USING SIGNON MANAGER	25
3.1 Introduction to Signon Manager	25
What is Signon Manager?	25
Access level indications	26
What Signon Manager provides	27
3.2 Signon Manager procedures	28
To sign on to a node	28
To override a signed-on user	30
To end an override of a signed-on user	32
To sign off a signed-on user	32
3.3 System perspective of Signon Manager	33
How Signon Manager works	33
Signon Manager scenarios	34
3.4 Signon Manager Configuration page	35
Configuration overview	35
Options tab	37

Contents

KeyLevel tab.....	40
Timeouts tab.....	42
3.5 Using the Signon Manager interface	45
Overview	45
Using the Signon bar	45
Using the Signon dialog box.....	49
Using the Signon taskbar icon.....	54
 4. LOADING THE LCN/TPN PERSONALITY.....	 55
4.1 Loading the Personality for an Operator Station	55
Purpose.....	55
LCNP Auto restart	55
Procedure to load a Personality into a GUS node	57
Manual load of a GUS node Personality (Non Local)	58
Auto Load of a GUS node Personality (Local).....	59
4.2 Loading the Personality for an APP node.....	60
Check to see if the Personality is already running.....	60
To load a personality into an APP node	60
Manual load of an APP node Personality	63
Auto load of an APP node Personality (Local)	64
Manual load of an APP node with AM Personality	65
Auto load of an APP node with AM Personality.....	66
 5. SHUTTING DOWN THE LCN/TPN PERSONALITY	 67
5.1 Shutdown a Node's LCN/TPN Personality	67
Use site-specific procedure	67
Procedure prerequisites	67
Shut down Windows only when necessary	68
 6. GENERATING LCN REVISION AND VERSION REPORTS.....	 69
6.1 Overview of LCN Revision and Version Reports	69
About LVRLOG	69
Accessing the online help.....	69
Installing LVRLOG application.....	70
Before starting the LVRLOG application	70
6.2 Using LVRLOG application.....	71
Syntax of LVRLOG command	71
Initiating LVRLOG application	73
Exiting LVRLOG application.....	74
Using LVRLOG application in user applications.....	74
Using LVRLOG application in batch files.....	75

6.3	LVRLOG application events and error messages	76
	LVRLOG application usage scenarios	76
	LVRLOG application events	76
	LVRLOG application error messages.....	82
7.	USING THE GUS LOGIC TOOL	85
7.1	Overview	85
	About the GUS Logic tool.....	85
	Configuring the xPM Logic display	85
	Prerequisites for using the xPM Logic display	86
7.2	Using the xPM Logic display	86
	Calling up the xPM Logic display	86
	xPM Logic display operations.....	89
	Elements of an xPM Logic display	91
7.3	Troubleshooting the xPM Logic display.....	95
	Error messages.....	95
	Display area does not indicate any values	95
	Communication error with LCN	95
	Tag not found	96
8.	VIEWING AND ANALYZING LCN LOAD	97
8.1	LCNLOAD Display	97
	Overview	97
	Calculating LCN Load	98
	Load Calculation Formula	98
	Invoking LCNLOAD display.....	99
	LCNLOAD display Operation	100
	Node load details	101
	LCN load details.....	101
8.2	Warnings and Error indications.....	102
	Overview	102
	Node-specific warning and error indications.....	102
	LCN warning and error indications	103
	List of messages and default limits	103
8.3	Collecting Historical data for LCN Load	104
	Overview	104
	Setting history collection rate	104
	Enabling/Disabling history collection	105
	Accessing historical LCN Load details	106

Tables

Table 1 Default TPS User Accounts	21
Table 2 LVRLOG application usage scenarios	76
Table 3 Event Description	78
Table 4 LVRLOG Error Description.....	82

Figures

Figure 1 TPS Node Startup Flowchart	17
Figure 2 Typical System Status Display	18
Figure 3 Signon Manager on GUS.....	33
Figure 4 xPM Logic display interface	90
Figure 5 Elements of an xPM Logic display.....	91

1. Introduction

1.1 Node fundamentals

LCNP board

A node containing an LCN Processor (LCNP) board is able to access the Local Control Network (LCN). Nodes that have an LCNP board include the following:

Operator Stations:

- Global User Station (GUS)
- Experion Station-TPS (ES-T)

Application/Database Servers:

- Application Processing Platform (APP)
- Experion Server-TPS (ESVT)
- Experion ACE-TPS (ACE-T)



ATTENTION

The LCN is also referred to as the TPN (TotalPlant Process Network).

Native Window

The Native Window is a user interface to the LCN/TPN, including access to operations, engineering, and maintenance functions.

The Native Window provides the functionality of the legacy Universal Station (US), within a PC window.

The following operator stations have the Native Window functionality:

- Global User Station (GUS)
- Experion Station-TPS (ES-T)

The GUS Native Window may connect to a locally installed LCNP board or to a different LCNP board in a remotely located GUS node. The Native Window running on the ES-T supports only local LCNP connections.

1. Introduction

1.1. Node fundamentals

Node startup and log on

Once power is applied to a node, the Microsoft Windows operating system starts up. This may be all that is necessary for the node to become operational, as the node may be configured to do an LCNP Auto Restart. LCNP Auto Restart automatically loads the necessary LCN/TPN personality software into the node's LCNP board.

After the power-on sequence, in order to operate the node, you must log on to Windows. After logging on to the node, check the System Status display to determine if the node's LCN/TPN personality needs to be loaded.

If the node did not do an LCNP Auto Restart, then the System Status Display will indicate that you must manually load the LCN/TPN personality.

Your system administrator can inform you as to whether your nodes are configured to do an LCNP Auto Restart.

Access level and Signon Manager

The operator station nodes (GUS and ES-T) provide a Signon Manager application that allows you to change your TPS system access level without logging off Windows.



ATTENTION

Signon Manager is required for a GUS node and/or ES-T and optional for Experion nodes.

In addition to Native Window, other applications also use Signon Manager to determine a user's authorization level. Applications that use Signon Manager are referred to as "Signon-aware" applications.

When you initially log on to Windows, Signon Manager is set to the same user account as Windows. If you log on to Windows as a user account that does not have authorization to manually load nodes, then, in order to load a node, you would need to sign on to Signon Manager with a user account that does have authorization.

TPS managed components

TPS nodes may have "managed" components that startup automatically when the node starts up. Managed components are components that, once configured, automatically appear on the System Management Display. The System Management display indicates the status of the node's components and it can be used to start the components, when necessary. The TPN Server is an example of a managed HCI (Honeywell Communication Interface) component.

With TPS R410, a system alarm is generated whenever an HCI software component (running on Windows environment) fails in an APP/GUS/ES-T/ESVT/ACE-T node. This feature is enabled by adding an auxiliary status for the GUS/APP/ES-T/ESVT/ACE-T nodes in the TPN network. This alarm is generated optionally for the failure of the HCI components. The composite HCI status is reflected in the auxiliary status of the node. When the auxiliary status changes to SEVERE, an alarm is annunciated and recorded in the System Status Journal.

This alarm is enabled by selecting the TPN Communication Service and the HCI State Notification target provided as part of the Configuration Utility.

The System Management feature helps to determine the HCI component status using the node composite status derived from Honeywell System Management Runtime. The HCI composite state is computed based on the state of the individual HCI components such as TPN Server, CL Server, Redirection Manager and so on.

When the composite status of the HCI components changes to WARNING, the status of the auxiliary node also changes to WARNING. If any of the HCI components is in the FAILED state, the status of the auxiliary node changes to SEVERE.



ATTENTION

Running TPN Communication Service is mandatory for monitoring the HCI components from TPN.

Troubleshooting: Error handling/Error messages

The following tables include a compilation of common error messages, the probable causes and the recommendations for resolution.

Scenario I

Description	TPN Communication Service fails to establish a TPS-TPN communication mechanism.
Cause	There is a communication error between TPN/TPS.
Effect	In the Application Log of Windows event viewer, an event is logged.
Recommendation/Resolution	Restart the TPN/TPS Communication Service.

1. Introduction

1.1. Node fundamentals

Scenario II

Description	TPN Communication Service is not responding (hangs/crashes).
Cause	There is a communication error between TPN/TPS.
Effect	A System Error journal entry is generated. The auxiliary status of the node is set to SEVERE with the AUX node text "Communication Error".
Recommendation/Resolution	Restart the TPN/TPS Communication Service.

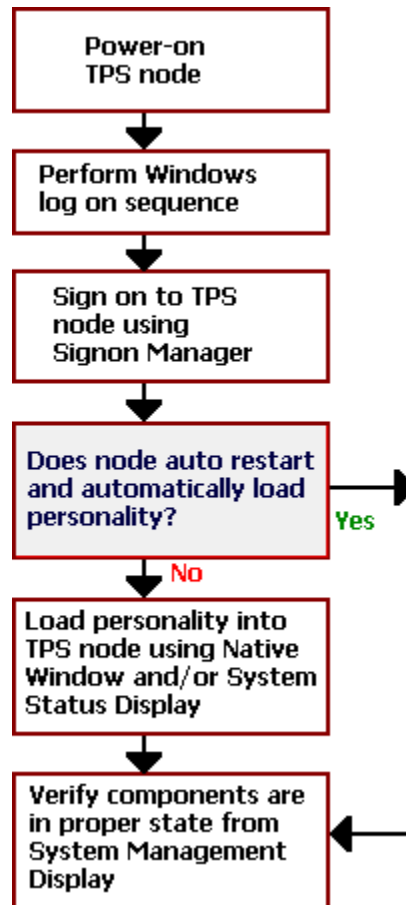
Scenario III

Description	TPS-TPN communication mechanism allocated for HCI State is not responding.
Cause	There is a communication error in the allocated channel.
Effect	A System Error journal entry is generated. The auxiliary status of the node is set to SEVERE with the AUX node text "Communication Error".
Resolution	Restart the TPN/TPS Communication Service.

TPS node startup flowchart

The following figure shows a flowchart of the general steps involved in starting up a TPS node (GUS or APP). The term "TPS node" does NOT refer to LCN-connected Experion nodes.

Figure 1 TPS Node Startup Flowchart

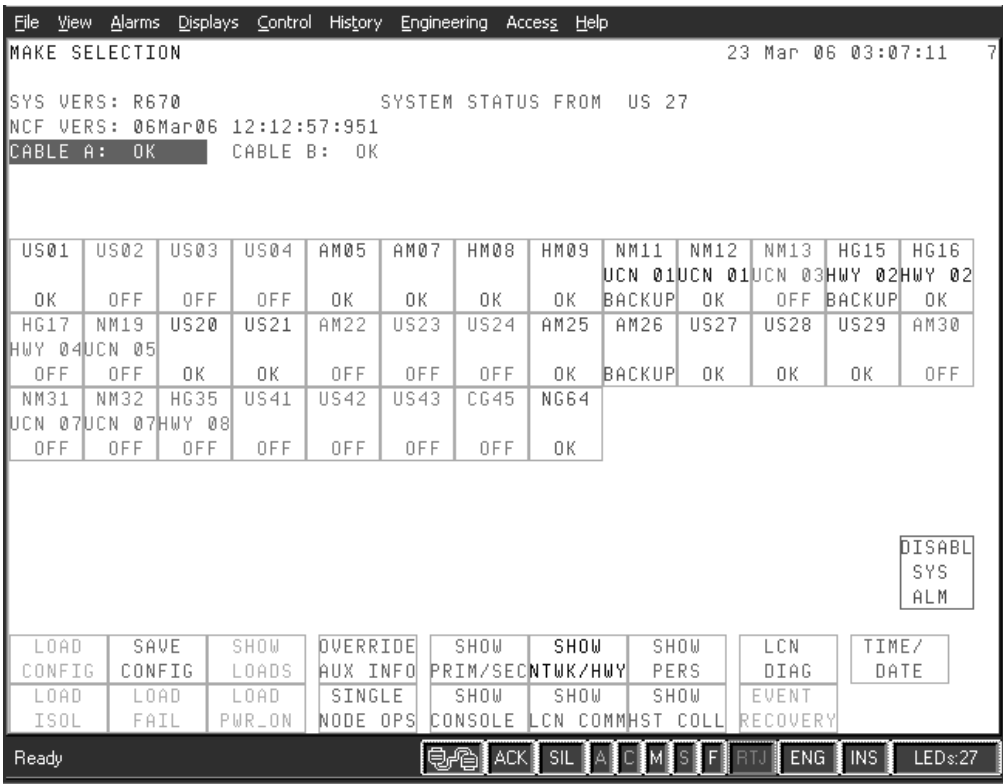


1.2 Status displays

System Status display

The following figure shows a typical System Status display of nodes connected to the LCN/TPN. It can be displayed in the Native Window and can be used to load nodes, such as APP nodes.

Figure 2 Typical System Status Display



REFERENCE - INTERNAL

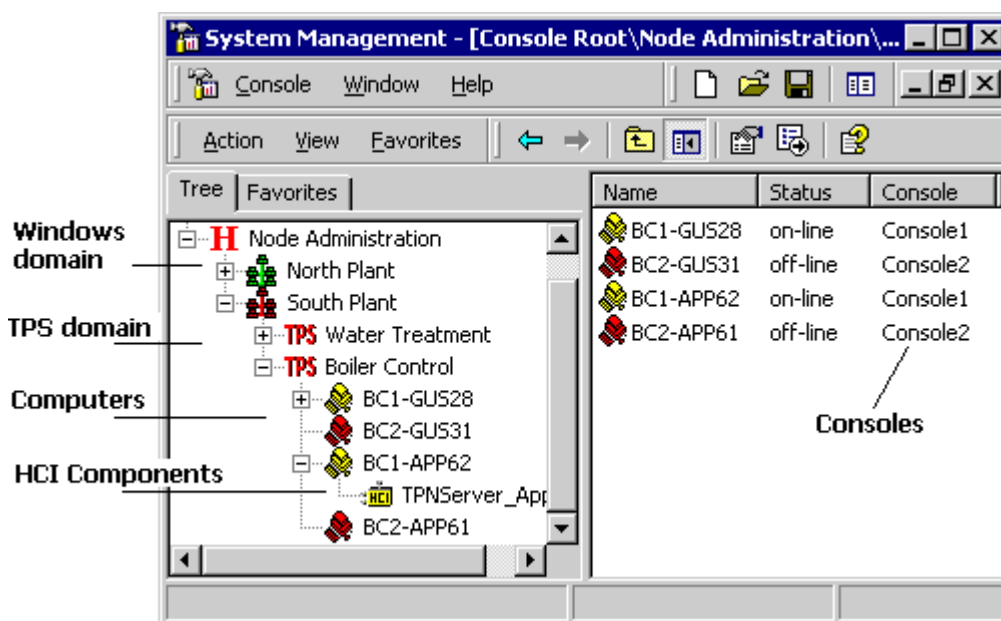
For details on the LCN/TPN System Status Display, see Section 11 of the LCN/TPN Process Operations Manual.

System Management display

The System Management Display provides additional status about system nodes. The display provides a hierarchical view of Windows domains, TPS domain Organizational Units (OU), system nodes (computers), and each node's managed components.

Computers organized into TPS domain OUs may also indicate their console assignment, in the right pane of the display.

The following figure shows a typical **System Management** display that appears within a Microsoft Management Console (MMC).



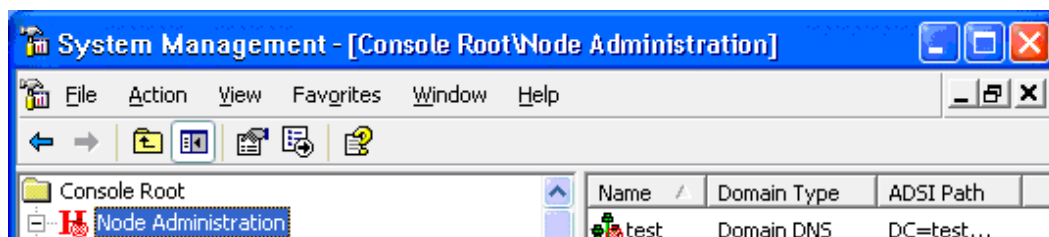
REFERENCE - INTERNAL

For details on the System Management Display, see the "System Management Display Organization" section of the *System Management Operations Guide*.

1. Introduction

1.2. Status displays

The following figure shows an MMC running on a Windows XP system that appears with a single menu bar where the File menu replaces the Console menu. System Management display operations remain the same for both Windows 2000 and Windows XP systems.



2. Logging on to a TPS node



ATTENTION

This section does not apply to Experion nodes. For logon information that applies to Experion nodes, see the *Experion Operators Guide*.

2.1 Logon Requirements for TPS nodes

When logging on to a TPS node, you are required to provide a user name and password. The following user accounts are configured for use in a TPS system:

Table 1 Default TPS User Accounts

User Name	Password
Administrator ¹	Configured by your TPS System Administrator
Operator ^{2,3}	Configured by your TPS System Administrator
Supervisor ^{2,3}	Configured by your TPS System Administrator
Engineer ²	Configured by your TPS System Administrator
TPSAdministrator ²	Configured by your TPS System Administrator

Notes:

1. The Administrator account is a local account used for installing software.
2. These accounts are Windows network accounts to be used when the TPS node is connected to a network.
3. Typically, the Operator and Supervisor accounts are disabled on APP nodes. When access to the System Management Display is required at the APP node, to view any other application, these accounts may be enabled.

Custom TPS user accounts

Your system administrator may provide additional user accounts, and can provide the necessary information about those accounts.

2.2 TPS node log on procedures

Logon procedures

There are three procedures for logging on to a TPS node:

- Logging on to a stand-alone TPS node (not part of a Windows domain network)
- Logging on to a networked TPS node (part of a Windows domain network)
- Locally logging on to a networked TPS node (network unavailable)

To log on to a stand-alone TPS node

Step	Action
1	Press CTRL+ALT+DELETE .
2	In the Logon dialog box, a) Type administrator in the User Name field. b) Type the password into the Password field.
3	Click OK .

To log on to a networked TPS node (network is available)

Step	Action
1	Press CTRL+ALT+DELETE .
2	In the Logon dialog box, verify that there is an entry for Domain . If there is no entry, stop here and do the next procedure to log on locally.
3	In the Logon dialog box, a) Type the appropriate user name in the User Name field. The user name is available from your TPS system Administrator or from Table 1, "Default TPS User Accounts." b) Type the password in the Password field.
4	Verify that the entry in the Domain field is set to the correct logon domain. Your TPS system administrator can provide you this information.
5	Click OK .

2. Logging on to a TPS node

2.2. TPS node log on procedures

Step	Action
6	If the message, "Can't locate Roaming Profile, do you wish to use the locally cached profile" appears, click Yes .
7	If the message, "Locally cached profile is newer than your Roaming Profile, do you wish to use the locally cached profile" appears, click No .
8	If the message, "Can't find user" appears, you must confirm the User Name, Domain, and Password. If they are correct, you must do the next procedure to log on locally.

To log on locally to a networked TPS node (network is not available)

Step	Action
1	In the Logon dialog box, a) Change the entry in the Domain field to the name of the local computer. b) Type administrator in the User Name field. c) Type the password in the Password field. d) Click OK.
2	Remember that the local logon account is being used. When access to the network is restored, log off and then log on to the node as a domain user.

2. Logging on to a TPS node

2.2. TPS node log on procedures

3. Using Signon Manager

3.1 Introduction to Signon Manager

What is Signon Manager?



ATTENTION

Signon Manager is required for GUS nodes and/or ES-T nodes but optional for Experion nodes.

- For GUS nodes, Signon Manager is installed when you install the TPS Base software.
 - For Experion nodes, Signon Manager is a separate package that you can install from the Experion DVD.
-

Signon Manager allows client applications to do security checks on the current system user (other than the current Windows user) without requiring a full Windows log off and log on, which would result in a brief loss of view at that workstation. This, for example, enables a node to auto-logon at boot time. The auto-logged user is a generic operations account with minimum permissions.

When the current user uses Signon Manager to sign on to a station, the Signon-aware applications grant access based on that user's permissions. During the shift, if a process adjustment is required that is not within that user's permissions, a supervisor can temporarily override the user's access to make the adjustment, and then end the override to restore the user's access to the station.

In addition, Signon Manager is the source of the LCN/TPN keyswitch state for the node and is a software version of the Integrated Keyboard (IKB) physical keyswitch.

3. Using Signon Manager

3.1. Introduction to Signon Manager

Access level indications

CAUTION

Indications of the keylock access level may be different between the Native Window, GUS pictures, and the physical keyswitch. You should use the indication in the status bar of the GUS or Native Window to determine the access currently in effect for that particular window.

With TPS R420/TPN R682 and later, the access levels in the Native Window and the TPN is synchronized, only if the login is changed using one of the following:

- Signon Manager login
- Windows login
- IKB Keyswitch
- Status bar buttons

The access level applied is based on the Signon Manager Configuration page settings.

What Signon Manager provides

Signon Manager has the following features:

- Eliminates the use of shared passwords.
- Provides a method for logging on to the system when the workstation user changes.
- Uses Windows security and user IDs as a basis for system security.
- Prevents loss of view during a change in authorization.
- Eliminates unexpected changes in access through GUS pictures and/or Native Window when either or both is remote.

3.2 Signon Manager procedures

This section describes the following Signon Manager procedures:

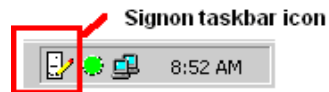
- Signing on to a node
- Overriding a signed-on user
- Ending an override of a signed-on user
- Signing off a signed-on user

To sign on to a node

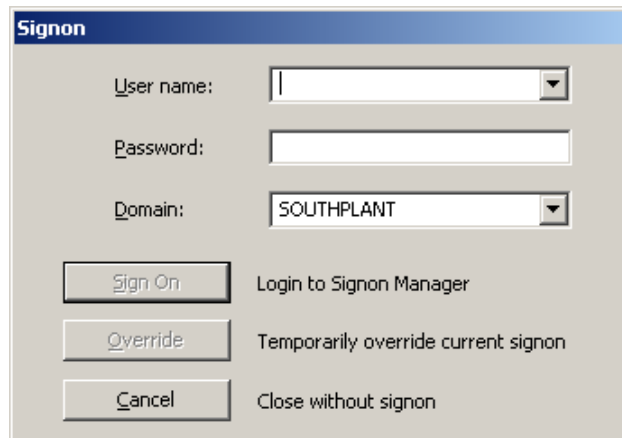
Step	Action
1	Click Sign On on the Signon bar,



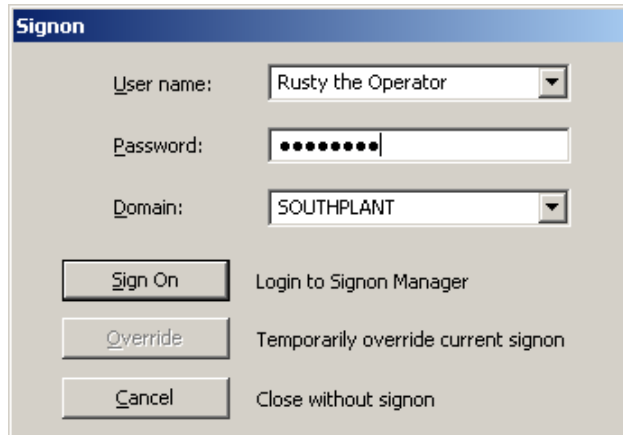
or click the Signon taskbar icon.



Result: The following **Signon** dialog box appears.

A dialog box titled "Signon". It contains three input fields: "User name:" with a dropdown arrow, "Password:" with a text box, and "Domain:" with a dropdown arrow showing "SOUTHPLANT". Below these fields are three buttons: "Sign On", "Override", and "Cancel". To the right of each button is a description: "Login to Signon Manager" for "Sign On", "Temporarily override current signon" for "Override", and "Close without signon" for "Cancel".

Step	Action
2	Type a valid user name and password. Result: The Sign On button is enabled.

A screenshot of the 'Signon' dialog box. It has a blue title bar with the text 'Signon'. Inside, there are three input fields: 'User name:' with a dropdown menu showing 'Rusty the Operator', 'Password:' with a masked password of ten dots, and 'Domain:' with a dropdown menu showing 'SOUTHPLANT'. Below these fields are three buttons: 'Sign On', 'Override', and 'Cancel'. To the right of each button is a description: 'Login to Signon Manager' for 'Sign On', 'Temporarily override current signon' for 'Override', and 'Close without signon' for 'Cancel'.

- 3 Click **Sign On**.
Result: On the Signon bar, the name of the signed-on user is displayed and the **Sign Off** button is enabled.

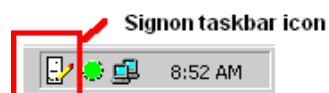


3. Using Signon Manager

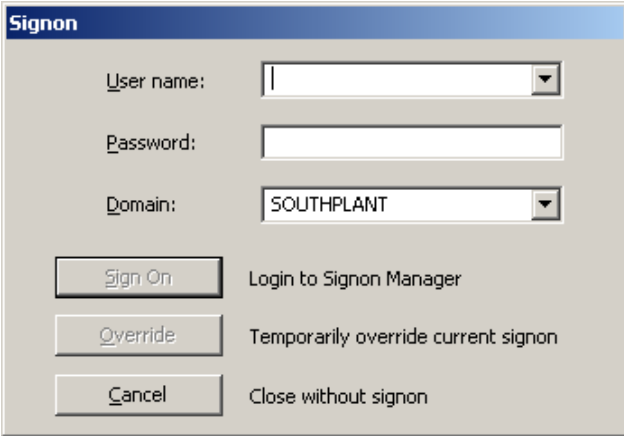
3.2. Signon Manager procedures

To override a signed-on user

Step	Action
1	Click Sign On/Override on the Signon bar,  or click the Signon taskbar icon.



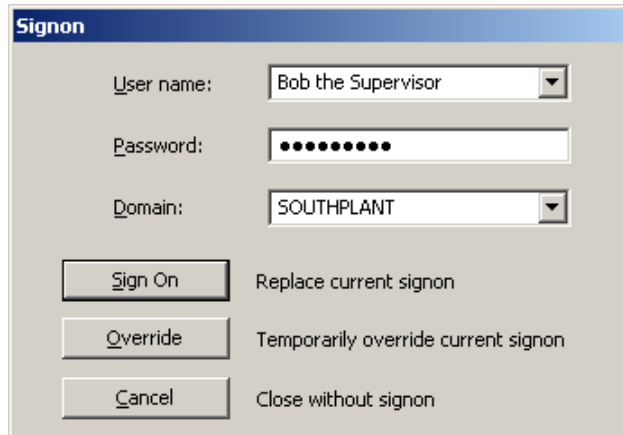
Result: The following **Signon** dialog box appears.



The Signon dialog box contains the following fields and buttons:

- User name:** A text input field with a dropdown arrow.
- Password:** A text input field.
- Domain:** A text input field with a dropdown arrow, currently showing "SOUTHPLANT".
- Buttons:**
 - Sign On:** Login to Signon Manager
 - Override:** Temporarily override current signon
 - Cancel:** Close without signon

Step	Action
2	Type a valid user name and password. Result: The Sign On and Override buttons are enabled.

A screenshot of the 'Signon' dialog box. It has a title bar 'Signon' in blue. Inside, there are three input fields: 'User name:' with a dropdown menu showing 'Bob the Supervisor', 'Password:' with a masked field of dots, and 'Domain:' with a dropdown menu showing 'SOUTHPLANT'. Below these are three buttons: 'Sign On' (with a tooltip 'Replace current signon'), 'Override' (with a tooltip 'Temporarily override current signon'), and 'Cancel' (with a tooltip 'Close without signon').

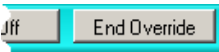
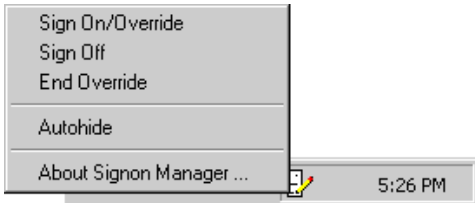
- 3 Click **Override**.
Result: On the Signon bar, the name of the override user is displayed and the **End Override** button is enabled.



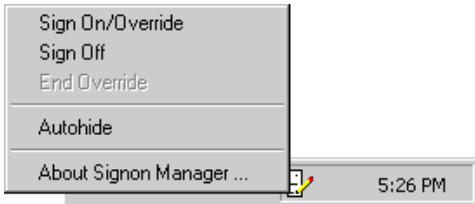
3. Using Signon Manager

3.2. Signon Manager procedures

To end an override of a signed-on user

Step	Action
1	Click End Override on the Signon bar,  or right-click the Signon taskbar icon and then click End Override .  Result: The Signon bar returns to the previous Signon display. 

To sign off a signed-on user

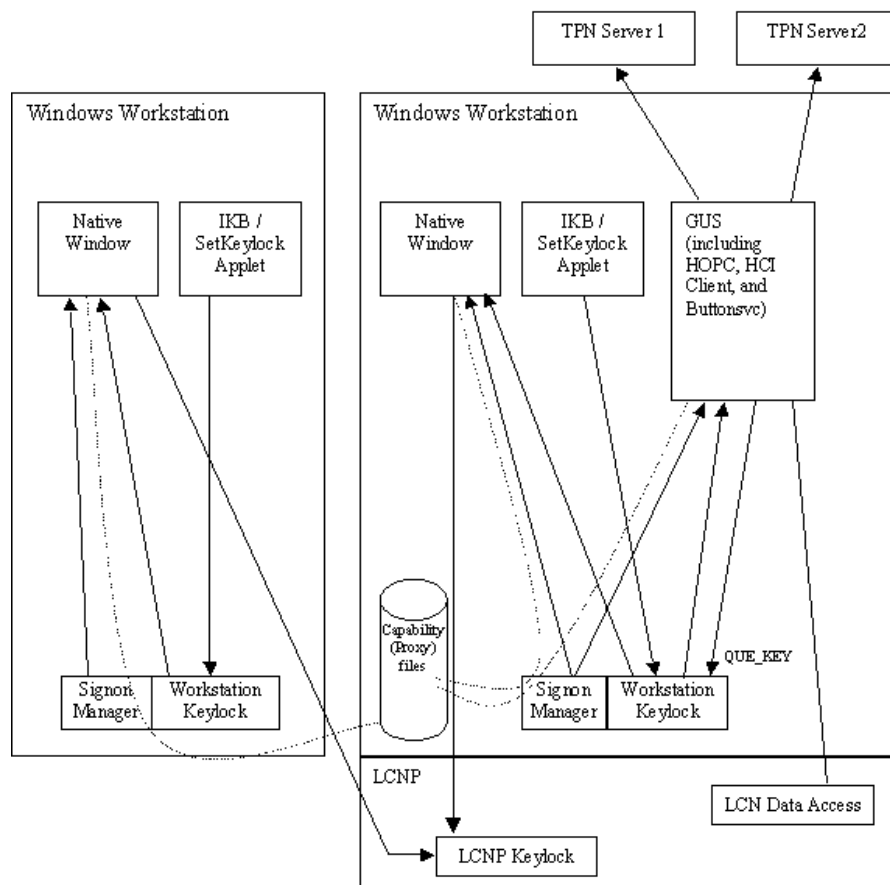
Step	Action
1	Click Sign Off on the Signon bar,  or right-click the Signon taskbar icon and then click Sign Off .  Result: The Signon bar returns to the default display. 

3.3 System perspective of Signon Manager

How Signon Manager works

The following figure shows how Signon Manager works within the context of a TPS system. The examples used for Signon-aware applications are Native Window and GUS displays. In the example, Native Window is shown running either local or remote while GUS is shown as running Local Only. Remoting would apply in the same way as it does to Native Window. All applications interact only with the Signon Manager and Workstation Keylock of their own node.

Figure 3 Signon Manager on GUS



3. Using Signon Manager

3.3. System perspective of Signon Manager

Access to GUS data, whether through a HOPC or TPN Server, is not dependant on the state of the LCNP keylock—it maintains its own keylock state, which is driven by the Workstation Keylock and GUS scripting commands. This eliminates the undesirable keylock changes that could result from various combinations of local and remote GUS pictures and Native Window.

Signon Manager scenarios

User login to system

When a user logs on to Windows on a system running Signon Manager, the Signon bar that appears shows the user's login ID. At this point, any security based on Signon Manager validation will use the logged-on user's ID—the same is true after a user signs off.

Application start

When a user starts a Signon-aware application such as Native Window, that application registers with Signon Manager. Signon Manager's immediate response to the registration is to send an event to the new registered application notifying it that there is a "new" user and a "new" keyswitch setting. This event triggers the application to validate the current Signon user (or Windows user if there are no Signons) against whatever criteria it is configured to use. In the Native Window case, this is a set of proxy files located on the system of the LCNP where it is connected. Site administrators configure these proxy files to define user or group access permissions to the LCN/TPN.

Sign on to Signon Manager

When a new user signs in, Signon Manager notifies all registered applications of the occurrence. This notification triggers the application to validate the new user's permission to the current security level in use by that application. If the level is higher than the new user is allowed, the level is automatically lowered to an acceptable level. If the current level is lower than the new user is allowed, the application may be configured to automatically raise the level to the highest allowed to the new user.

Change of connection

User permissions may differ according to the data source to which a TPS application connects. When a TPS application changes or adds a connection, it must use Signon Manager to validate and adjust access, based on the currently signed-on user.

3.4 Signon Manager Configuration page

Configuration overview

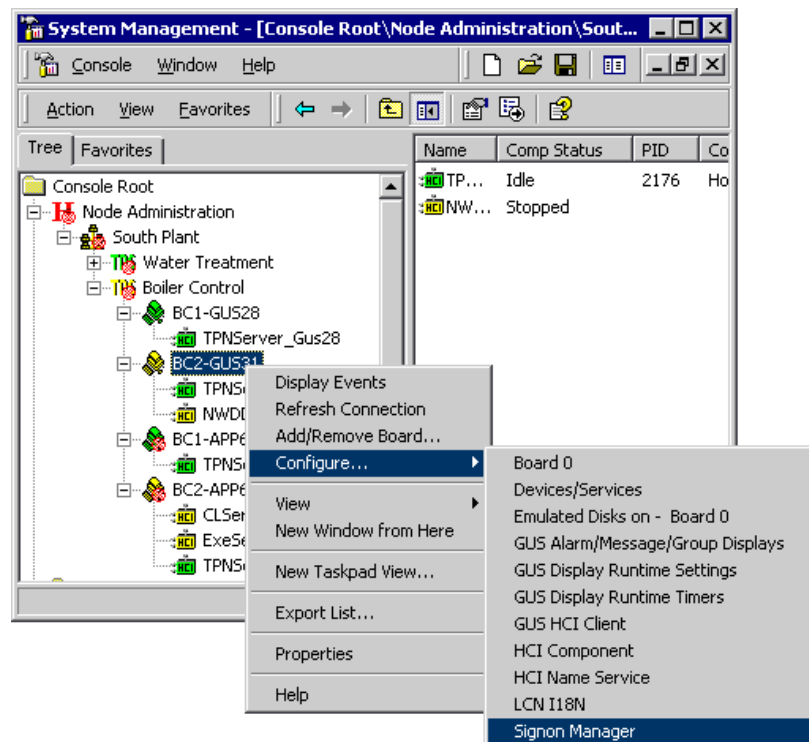


ATTENTION

This section refers to TPS nodes only. For information about configuring Signon Manager for Experion nodes, see the "Using Integrated Security" topic in the *Experion Server and Client Configuration Guide*.

This section is intended for users who configure Signon Manager behaviors. It describes the options and suggested settings for the Signon Manager. The following examples show the defaults that are used if you do not change the configuration before the first use.

You set the configuration parameters for the Signon Manager through the Configuration Utility, which is accessible through **Start > Programs** or from the **System Management** display as shown in the following figure:



3. Using Signon Manager

3.4. Signon Manager Configuration page

The resulting **Signon Manager** page provides options for configuring its display behavior. Any changes made through this page become effective immediately when you click **Apply** (page remains open) or **OK** (page closes).

The **Signon Manager** page includes the **Options**, **KeyLevel**, and **Timeouts** tabs as described in the following paragraphs. Note that the **Apply** button is initially disabled until you change one or more configuration settings from the last saved value on any tab.

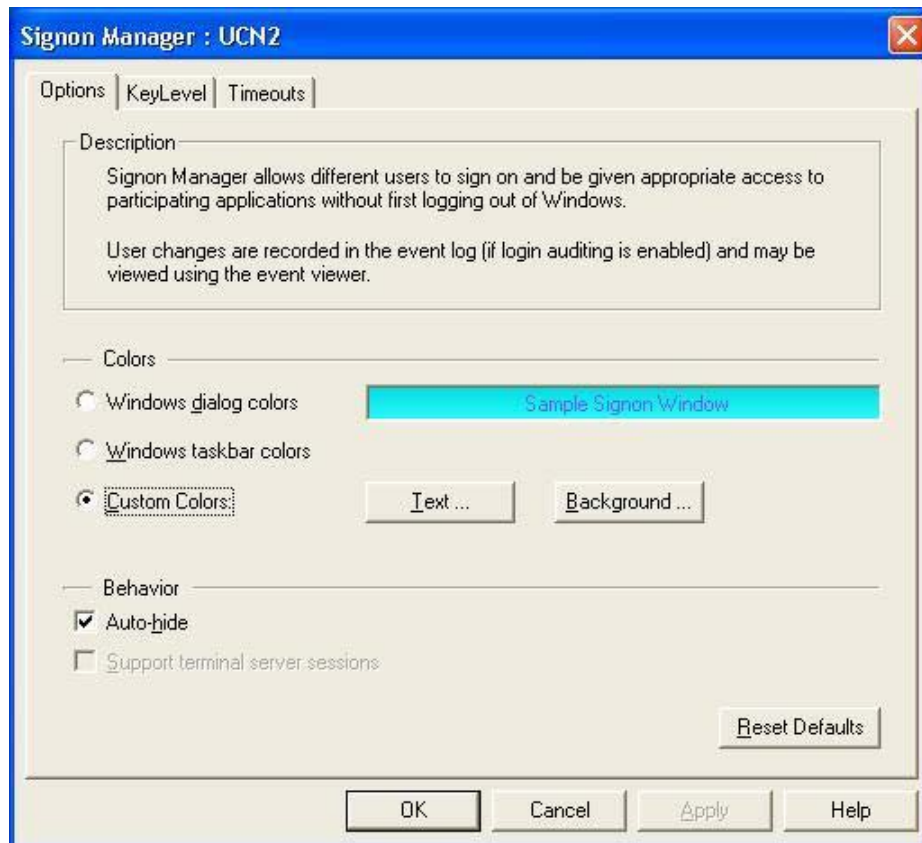


TIP

Signon event logging (along with normal logon/logoff events) can be configured using Windows administration tools. Signon events are logged in the Windows security event log. However, the security event log will eventually overwrite itself. Thus, to create a permanent record, you must use a third-party tool to capture the event log.

Options tab

The following **Options** tab provides options for Signon Manager display preferences:



3. Using Signon Manager

3.4. Signon Manager Configuration page

Colors

The text and background colors of the Signon Manager display can be configured to make it more or less obvious, depending on site preference. The color schemes for the Signon Manager display are defined by the Windows display properties (accessible through *Start > Settings > Control Panel > Display Properties > Appearance*). The default is **Customize Colors** with blue text on a cyan (light blue-green) background.

- **Window dialog Colors** sets the Signon bar color scheme to resemble a Windows dialog box.
- **Window taskbar Colors** sets the Signon bar color scheme to resemble a Windows taskbar. If “Windows Classic style” is selected on the *Display Properties > Appearance* tab, this setting behaves the same as **Dialog Colors**. If “Windows XP style” is selected, the colors are selected by the *Color scheme* setting.
- **Custom Colors** uses the standard Windows color selection dialog box to select text and background colors.
- **Text** and **Background** displays the standard Windows color selection dialog box.

Behavior

- **Auto-Hide**

By default, this check box is selected, causing the Signon bar to remain hidden until you move the cursor to the edge of the display occupied by the Signon bar. Note that high priority windows can cover the small area of the bar that remains visible when the bar is auto-hidden, preventing the bar from becoming visible when the mouse moves to that display edge.

- **Support terminal server sessions**

When this option is set, each session has its own Signon bar, thus allowing Signon-aware applications within that session to adopt the permission assigned by Signon Manager for that session. If this check box is cleared, only session “0” (the display attached to the PC) has a Signon bar. All other sessions operate in the permission of the user logged on to that session, independent of Signon Manager.

Reset Defaults

This button restores all settings on this tab to the default values previously stated.

Apply

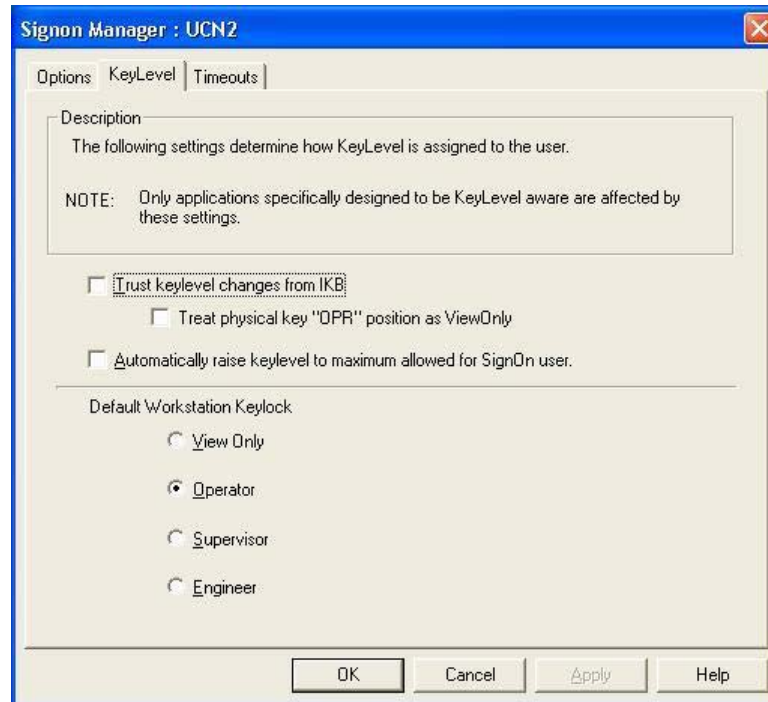
This button is enabled only if one or more configuration settings on any of the pages have changed from the last saved value.

3. Using Signon Manager

3.4. Signon Manager Configuration page

KeyLevel tab

The following **KeyLevel** tab provides options for configuring keylevel settings:



Trust keylevel changes from IKB

By default, this check box is selected. When set, LCN/TPN keyswitch changes originating from a physical integrated keyboard (IKB) are used, even if the currently signed-on user is not authorized at this level. For sites not using the IKB, this option is not used. For sites that do use the IKB, this option allows for a quick keylevel change without the delay of entering a user ID and password.

Treat physical key "OPR" position as ViewOnly

This option determines whether GUS displays will respond to a change in the IKB key to "OPR" by switching to OPR or VIEW. For the Native Window, this behavior is controlled by the NCF.

With TPS R420/TPN R682 and later, this option determines the access levels (OPR and VIEW) applied to the Native Window displays based on the change in the IKB key.

The following table explains the access levels applied to Native Window, GUS displays, and TPN:

Note: Assume IKB key switch is in OPR.

Treat physical key "OPR" position as ViewOnly option	Access level applied to Native Window	Access level applied to GUS displays	Access level applied in TPN	CURRENT DEFAULT value in the ACCESS CHG target of CONSOLE STATUS
Selected	VIEW	VIEW	VIEW	VIEW
Cleared	OPR	OPR	OPR	OPR

Automatically raise keylevel to maximum allowed for Signon user

By default, this check box is cleared. When set, cooperating TPS applications automatically raise the LCN/TPN keylevel for a new Signon to the highest keylevel allowed for the new user. When this option is not set, TPS applications do not automatically raise the keylevel for a new Signon, but always lower the key setting if the new user is not authorized at the current level.

Default workstation Keylock

By default, this keylock is set to **View Only**. This allows a site to decide what the default keylevel will be for a workstation, if a physical keyswitch is not present. This default level is set whenever a Windows log on or log off occurs. This provides the starting point for the key setting for a workstation at the time of log on. If the auto-raise option is enabled, or a physical key switch is present, this setting is of little concern.

3. Using Signon Manager

3.4. Signon Manager Configuration page

Apply

This button is enabled only if one or more configuration settings on any of the pages have changed from the last saved value.

Timeouts tab

The following **Timeouts** tab provides two timeout options—*idle* and *override*. You can configure and enable each timeout independently.

The screenshot shows the 'Signon Manager : UCN2' window with the 'Timeouts' tab selected. The window has a blue title bar and a standard Windows-style interface. The 'Timeouts' tab is active, showing a description of the timeout features and several configuration options. The 'Description' box explains that Signon Manager supports two optional timeouts: idle and override. The 'Idle' timeout initiates a signoff when inactivity is exceeded, and the 'Override' timeout ends the override after a fixed period. Below the description, there are three checkboxes: 'Sign users off if there is no activity for:' (checked), 'Sign override users off after:' (unchecked), and 'User may select timeout of up to:' (unchecked). Each checkbox has a corresponding time value in minutes: 5, 10, and 60 respectively. A 'Timeout warning' section is also present, with a checkbox for 'Provide a timeout warning:' (checked) and a value of 30 seconds. There are also checkboxes for 'Show user interface if hidden' (unchecked) and 'Play sound:' (unchecked). At the bottom right of the configuration area are buttons for 'Browse...' and 'Reset Defaults'. The bottom of the window features a standard set of buttons: 'OK', 'Cancel', 'Apply', and 'Help'.

Signon Manager : UCN2

Options | KeyLevel | **Timeouts**

Description
Signon Manager supports two optional timeouts.
The idle timeout initiates a signoff when the configured period of inactivity is exceeded.
The override timeout will end the override after a fixed period of time from the override operation.

☒ Sign users off if there is no activity for: 5 minutes

☐ Sign override users off after: 10 minutes

☐ User may select timeout of up to: 60 minutes

Timeout warning

Provide a timeout warning: 30 seconds before signing users off

☐ Show user interface if hidden

☐ Play sound: [] Browse...

Reset Defaults

OK Cancel Apply Help

Sign users off if there is no activity for

This option when selected, tracks the time from the last user activity, which can be any keyboard, mouse, or touch screen activity. If the idle time exceeds the configured value, any existing sign on and override ends, thus returning to the default state. The default value is 5 minutes.

Sign override users off after

The override timeout is active only when an override user is signed on. The period is fixed, measured from the point of the override, and independent of system or user activity. When the timeout expires, the override ends without affecting the signed-on user and his/her security level. The default value is 10 minutes.

- User may select timeout up to:
 - This is the maximum period the user may select for the override timeout from the Signon dialog. The default value is 60 minutes.

Timeout warning

Signon Manager provides three warning mechanisms to notify a user of an impending idle or override timeout:

- The relevant timeout field on the Signon bar flashes (unconditional).
- The Signon bar automatically unhides, if hidden.
- A configured sound file plays.

3. Using Signon Manager

3.4. Signon Manager Configuration page

Although the first mechanism (flashing timeout field) is unconditional, you can configure the Signon bar and sound file as follows:

- **Provide a timeout warning:** This field, enabled when either timeout type is selected, determines the interval before a timeout when the warning mechanisms are activated (this interval is applicable to all three mechanisms). The default value is 30 seconds.
- **Show user interface if hidden:** By default, this check box is cleared. When selected, this option causes an auto-hidden Signon bar to automatically unhide whenever a timeout timer counts down to the interval previously selected.
- **Play sound:** By default, this check box is cleared. When selected, it enables the field for selecting a warning sound file. When selected, and a valid sound file is configured, Signon Manager will start playing the sound file when the timeout timer counts down to the interval previously selected.

Note: The warning sound volume is independently controlled by the user—not the Signon Manager.

Reset Defaults

This button restores all settings on this tab to the default values previously stated.

Apply

This button is enabled only if one or more configuration settings on any of the pages have changed from the last saved value.

3.5 Using the Signon Manager interface

Overview

The Signon Manager user interface consists of the *Signon bar*, *Signon dialog box*, and *Signon taskbar icon*, as described in the following paragraphs. When a user logs on, the user interface application starts automatically and continues until the user logs off.

Using the Signon bar

The Signon bar always displays the current user(s) signed on to Signon Manager to validate security in the various Signon-aware applications.

The Signon bar has the similar appearance and features of a Windows taskbar. You can configure it with a Windows color scheme or use custom colors to emphasize or de-emphasize it according to site policy. You can also configure it to auto-hide and/or dock to one edge of a display (the default position is at the top of the primary display).

The Signon bar has the *Always on Top* attribute. If it is hidden by another *Always on Top* window, you can access Signon Manager by clicking the Signon taskbar icon or pressing **CTRL+ALT+S**.

Shortcut menu

Right-clicking anywhere on the Signon bar (except on enabled buttons) displays the following shortcut menu:



3. Using Signon Manager

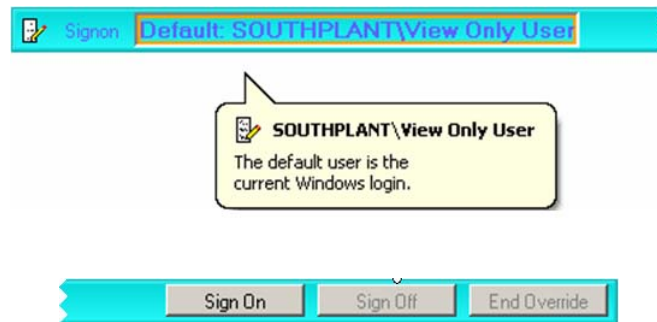
3.5. Using the Signon Manager interface

Signon bar states

The Signon bar appears in one of three states—*Default*, *Signon*, or *Override*—as described in the following three sections:

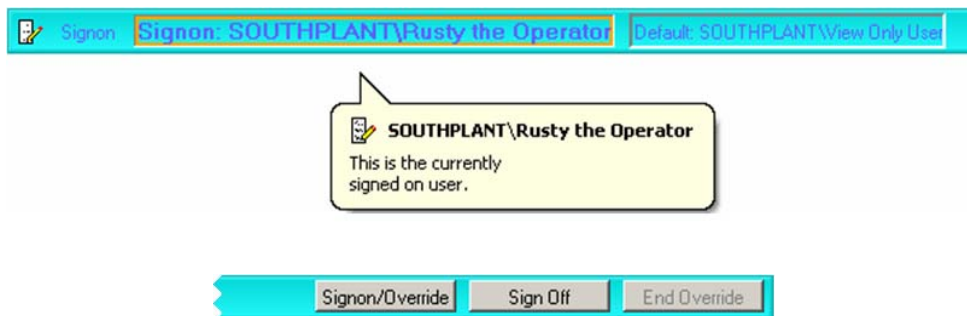
Default state

The following figure shows the default Signon bar when no user is signed on to the Signon Manager. The domain/user ID that appears is that of the currently logged-in interactive Windows user.



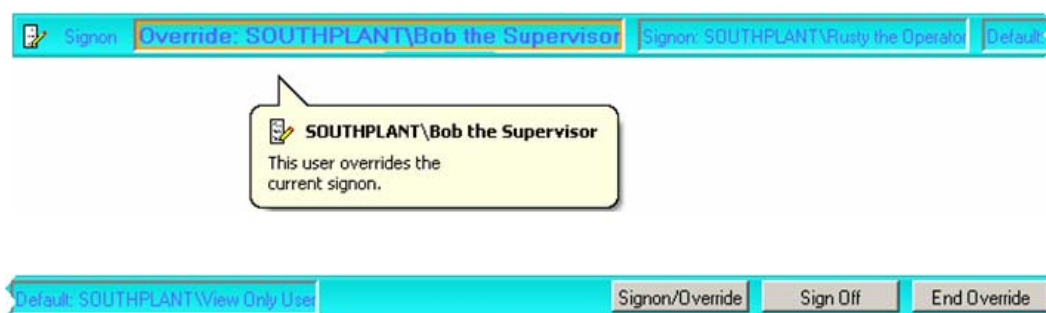
Signon state

The following figure shows the Signon bar after signing on to Signon Manager. The domain/user ID is the Windows user ID that Signon Manager has validated through the Signon dialog box (also shown), and the domain in which that ID is registered. Clicking the **Sign Off** button signs off the user and returns the Signon bar to the *default* state.



Override state

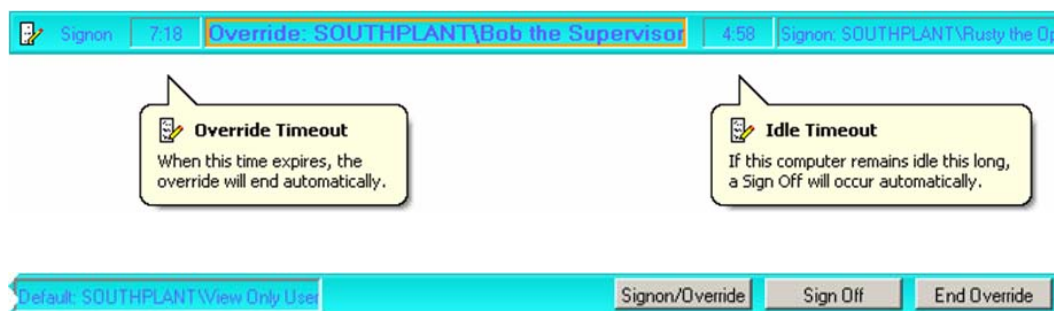
The following figure shows the Signon bar when the current sign on is temporarily overridden. Clicking the **End Override** button returns the Signon bar to the *Signon* state, and the user ID (used for validation) returns to the prior sign on. Clicking the **Sign Off** button ends both the override and the Signon, and the user ID returns to the default (Windows login).



Timeout fields

Signon Manager also supports two optional timeout conditions—*idle* and *override*—as described in the previous “Timeouts tab” topic.

The following figure shows each timeout field, along with its related tool tip that shows when the cursor hovers over the field. The idle timeout field (adjacent to the **Signon** field) shows only when an idle timeout is enabled and a signon is active. The override timeout field (adjacent to the **Override** field) shows only when an override timeout is enabled and an override is active.



3. Using Signon Manager

3.5. Using the Signon Manager interface

Auto-hide function

By default, the Signon bar auto-hides and docks at the top of the primary display. However, you can configure it to not auto-hide.

If the Signon bar is set to auto-hide (but not behind another *Always on Top* window), you can make it appear by moving the cursor to the docking edge of the display. To make it auto-hide again, move the cursor away from the docking edge.

Windows allows only one application taskbar (including the Windows taskbar) to auto-hide at a given edge of a display. Thus, if the Signon bar and Windows taskbar are both positioned at the same display edge, a conflict will occur, causing the Signon bar to automatically disable its auto-hide function—this is also true for a multi-display system.

When the Signon bar is auto-hidden, you can access it by one the following methods:

- Move the cursor to make the Signon bar show, and then click **Signon/Override**.
- Click the Signon taskbar icon.
- Use the **CTRL+ALT+S** hot key.

The following conditions will also make the Signon bar appear:

- Whenever the Signon dialog window shows.
- Whenever a timeout within the configured feedback interval is imminent and Unhide user interface if hidden is selected on the “Timeouts tab”.
- The Signon bar will also remain visible while a new Signon is being validated with Windows.

Signon bar and SafeView operation

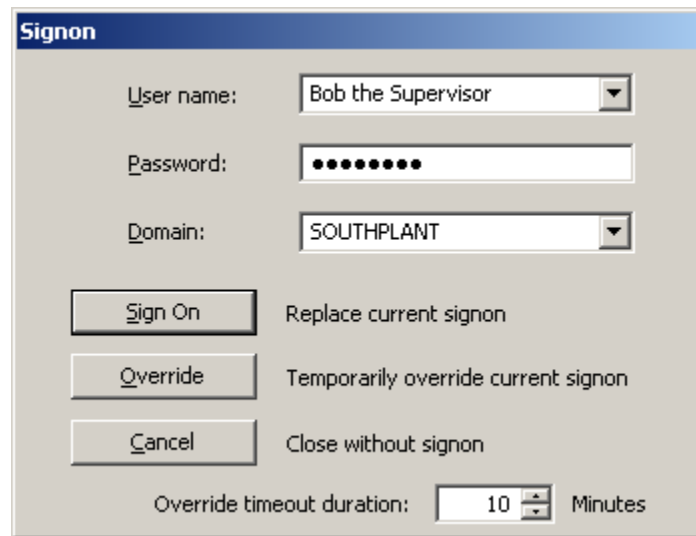
The Signon bar may affect the SafeView configuration for the system.

- If the Signon bar is set to auto-hide (by default), it does not normally occupy desktop space. Thus, SafeView can manage the entire display (less any space reserved for the Windows taskbar).
- If the Signon bar is not configured to auto-hide, it reserves sufficient desktop space for its display, thus reducing the overall area that SafeView can manage. For this condition, you may need to update the SafeView configuration to account for the Signon bar’s reserved space.

Using the Signon dialog box

The Signon dialog box allows a new user to sign on to Signon Manager. The Signon dialog box opens by clicking **Signon/Override** on the Signon bar or the Signon taskbar icon.

Prior to the first Windows session sign on, the Signon dialog box is blank. Depending on the signon state, Signon dialog boxes may display different text, and some buttons may be disabled. The following figure is an example of a Signon dialog box:



As with the Signon bar, the Signon dialog box also has the *Always on Top* attribute to ensure that normal application windows will not cover it. However, if another *Always on Top* window covers the Signon dialog box, you can bring it forward by clicking the Signon taskbar icon or pressing **CTRL+ALT+S**.

Depending on the current state of the display, pressing **CTRL+ALT+S** causes one of the following actions:

- If the Signon dialog box is opened, but off-screen or hidden by other windows, the hot key displays the Signon dialog box, centered on the primary display and on top of all other windows.
- If the Signon dialog box is not opened, the hot key opens the Signon dialog box, centered on the window where it is docked.

3. Using Signon Manager

3.5. Using the Signon Manager interface

Signon dialog box buttons and fields

The following table describes the controls and fields on a typical Signon dialog box:

Control or field	Description/function
User name and Domain fields	Show the values for the prior successful sign on or override.
User name drop-down arrow	Opens a list of prior successful user names.
Domain drop-down arrow	Opens a list of available domains.
Password field	Allows you to type the password.
Sign On button	Enabled when you type a user name in the User Name field.
Override button	- Validates the user and, if successful, updates the signed-on user as the new user. - Remembers the prior user (so that the end override action can return to that prior user).
Cancel button (or ESC key)	Cancels the sign on activity and removes the Signon dialog box from the display.
Override timeout duration/Minutes spin box	Displayed whenever Signon is enabled, <u>and</u> with the following Override Timeout settings: Override Timeout is enabled Allow user to adjust override timeout interval is selected The Minutes value defaults to the configured override timeout interval but is adjustable by the overriding user.

User validation

When you select a sign on or override action, Windows validates the current **User Name**, **Password**, and **Domain** entries to determine if:

- The user is registered.
- The user has permission to log on to the system. The system notifies the user of the results of the validation and allows the user to try again if the validation fails.

Signon dialog box states

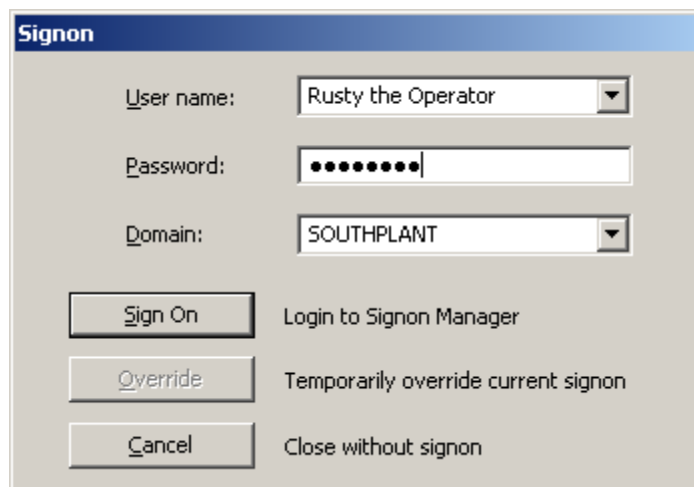
The Signon dialog box appears in one of three states—*Default*, *Signon*, or *Override*—as describes in the following three sections:

Default state

The following figure shows the Signon dialog box while Signon Manager is in the default state (no current signon). **Sign On** is the default selection.

Typing a user name enables **Sign On**, thus establishing the default state (that is, the selected action is done when you press the **ENTER** key). In this state, **Override** remains disabled because there is no current signon to override.

Clicking **Sign On** validates the **User name**, **Password**, and **Domain** entries. If all three entries are valid, Signon Manager establishes this user as the current signon, thus superseding the default user for security purposes.



3. Using Signon Manager

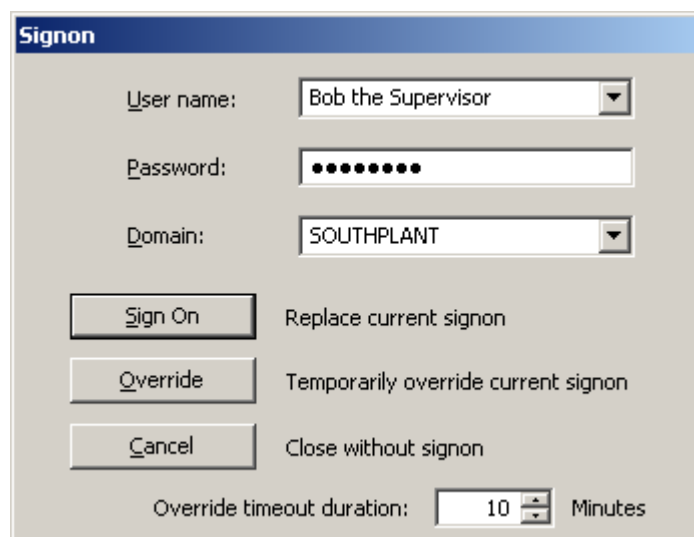
3.5. Using the Signon Manager interface

Signon state

The following figure shows the Signon dialog box when a user is signed on. The default selection is **Override**.

Clicking **Sign On** validates the user and, if successful, signs off the current user, and then signs on the new user.

Clicking **Override** validates the user and, if successful, establishes the user as the current override user, superseding the current Signon for security purposes until the override ends.



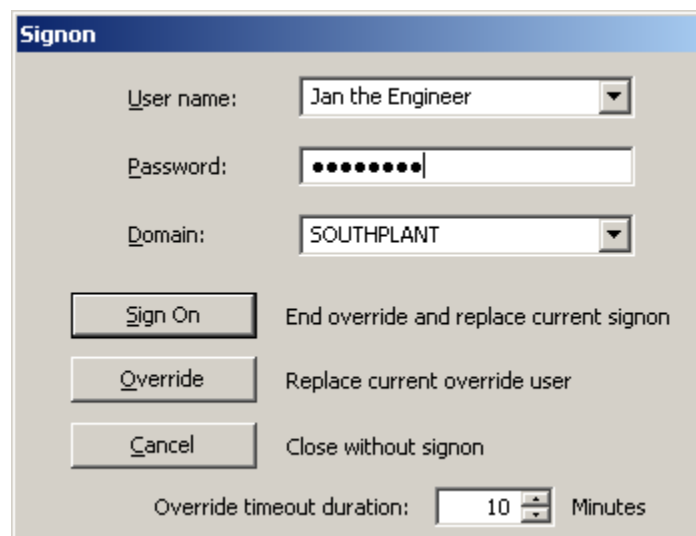
The image shows a Windows-style dialog box titled "Signon". It contains three input fields: "User name:" with a dropdown menu showing "Bob the Supervisor", "Password:" with a masked field of ten dots, and "Domain:" with a dropdown menu showing "SOUTHPLANT". Below these fields are three buttons: "Sign On", "Override", and "Cancel". To the right of each button is a description: "Replace current signon" for Sign On, "Temporarily override current signon" for Override, and "Close without signon" for Cancel. At the bottom of the dialog is a label "Override timeout duration:" followed by a spinner box set to "10" and the word "Minutes".

Override state

The following figure shows the Signon dialog box during an override. The default selection is **Override**.

Clicking **Sign On** validates the user and, if successful, signs off both the current user and the current override, and then signs on the new user.

Clicking **Override** validates the user and, if successful, replaces the current override user, thus establishing the new user for security purposes until the override ends.



The image shows a Windows-style dialog box titled "Signon". It contains the following elements:

- User name:** A text box with "Jan the Engineer" and a dropdown arrow.
- Password:** A text box with ten dots representing a masked password.
- Domain:** A text box with "SOUTHPLANT" and a dropdown arrow.
- Buttons:** Three buttons are arranged vertically: "Sign On", "Override", and "Cancel".
- Descriptions:** To the right of the buttons are their functions:
 - Next to "Sign On": "End override and replace current signon"
 - Next to "Override": "Replace current override user"
 - Next to "Cancel": "Close without signon"
- Timeout:** At the bottom, "Override timeout duration:" is followed by a spinner box set to "10" and the word "Minutes".

3. Using Signon Manager

3.5. Using the Signon Manager interface

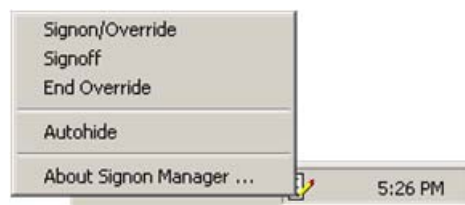
Using the Signon taskbar icon

The Signon taskbar icon provides access to the Signon Manager. Whenever the icon shows, the Signon Manager user interface is active, even if the Signon bar is hidden. In this case, clicking the icon is the only method of starting a new signon or a signoff.

Clicking the Signon taskbar icon opens the Signon dialog box. If the dialog box is already opened but hidden by another window, clicking the icon brings the dialog box forward and centers it on the display. Moving the cursor over the icon reveals the current signon state and signed-on user.

Shortcut menu

Right-clicking the Signon taskbar icon displays the following shortcut menu:



4. Loading the LCN/TPN Personality

4.1 Loading the Personality for an Operator Station

Purpose

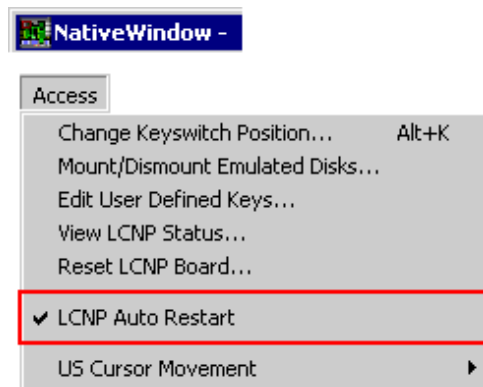
A software personality is required to make a node functional to communicate to the LCN (also referred to as the TPN). Once the LCN/TPN personality is loaded to an Operator Station, its Native Window becomes operational. The System Status Display can be displayed in the Native Window and used to load the LCN/TPN Personality of other nodes on the system.

LCNP Auto restart

Your system administrator may configure each node to do an LCNP Auto Restart, which automatically loads the node's LCN/TPN personality. Your administrator selects this configuration through the **Enable LCNP Auto Restart** checkbox on the LCNP Page of the node's Configuration Utility.

After configuring Enable LCNP Auto Restart, your system administrator may also add the menu item **LCNP Auto Restart** to the Native Window Access menu, providing the operator with the means to control (enable and disable) the LCNP Auto Restart function.

To determine if the operator has access to the LCNP Auto Restart function, select **Access** from the Native Window menu bar, as shown in the following figure:



4. Loading the LCN/TPN Personality

4.1. Loading the Personality for an Operator Station

If the **LCNP Auto Restart** menu item is checked, then whenever the node's LCNP board is in the PWR_ON state, the LCNP will be automatically loaded with the node's LCN/TPN personality. After the LCN/TPN personality is loaded, the other files such as the NCF (Network Configuration File), Area Database, and backplane Load Modules are loaded from the NET (network History Module).



ATTENTION

When LCNP Auto Restart is enabled, any time the LCNP goes into PWR_ON state, the LCNP will be automatically loaded.

When LCNP Auto Restart is enabled, you cannot perform a local/target (manual) load of the node. You must disable the LCNP Auto Restart feature before you can load a node manually.

The LCNP Auto Restart function does not apply to these node types:

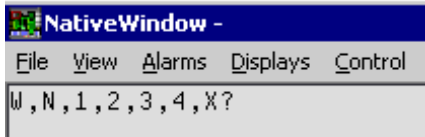
- Application Module Replacement (AMR)
 - Universal Station Replacement (USR)
-

4. Loading the LCN/TPN Personality

4.1. Loading the Personality for an Operator Station

Procedure to load a Personality into a GUS node

To load the Native Window (LCN/TPN personality) for an operator station, perform the following steps:

Step	Action
1	Click LOAD in the Native Window. 
	Result: The system prompt "W, N,1,2,3,4,X?" appears. 
	 ATTENTION Steps 2 and 4 that follow require a functional LCN/TPN History Module. If a History Module is not functional at the time of a node load, your system administrator may have a fast load cartridge configured to load the node. In this case, your system administrator can provide the response to give to the prompt presented in Steps 2 and 4.
2	If this node will operate as a <u>G</u> US, place the cursor in the Native Window and type W . (The personality image files will be loaded from the local hard disk). If this node will operate as a <u>U</u> S replacement, place the cursor in the Native Window and type N . (The personality image files will be loaded from the network History Module).
3	Press ENTER on the keyboard. Result: The node will begin loading its personality. A short time later, the following prompt appears in the Native Window: "NCF? N,1,2,3,4,X?"

4. Loading the LCN/TPN Personality

4.1. Loading the Personality for an Operator Station

Step	Action
4	Place the cursor into the window and type N .
5	Press ENTER on the keyboard. Result: The node loads the other files from the network History Module.
6	View the node in the Console Status Display of the Native Window. When the node's state in this display indicates OK , the node is available for use as part of the TPN.

Manual load of a GUS node Personality (Non Local)

The following table contains the procedure to load a GUS node with the GUS Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge (Non Local) - Manual Load:

Step	Action
1	Mount a fast-load cartridge that includes GUS personality files (directory &UXS) on a different GUS/US. Ensure that the fast-load cartridge does not include OPR personality files (directory &OPR).
2	From a US/GUS node on the LCN, press SYS STATS to open the System Status Display.
3	Select the required GUS node.
4	Select NODE STATUS .
5	Select the required GUS node.
6	Select LOAD/DUMP target.
7	Select MANUAL LOAD .
8	Select UNIVERSAL .
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number that contains Fast-load cartridge.

4. Loading the LCN/TPN Personality

4.1. Loading the Personality for an Operator Station

Step	Action
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the load is complete, the node status is displayed as OK , and the node type is displayed as GUS .

Auto Load of a GUS node Personality (Local)

The following table contains the procedure to load a GUS node with the GUS Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge (Non Local) – AutoLoad Local:

Step	Action
1	Mount a fast-load cartridge that includes GUS personality files (directory &UXS) on a different GUS/US. Ensure that the fast-load cartridge does not include OPR personality files (directory &OPR).
2	From a US/GUS node on the LCN, press SYS STATS to open the System Status Display.
3	Select the required GUS node.
4	Select NODE STATUS .
5	Select the required GUS node.
6	Select LOAD/DUMP target.
7	Select AUTOLOAD LOCAL .
8	Press ENTER to execute. Note: In some instances where the fast-load cartridge is mounted on the requesting node, it is not necessary to perform steps 9–15.
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .

4. Loading the LCN/TPN Personality

4.2. Loading the Personality for an APP node

Step	Action
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK , and the node type is displayed as GUS .

4.2 Loading the Personality for an APP node

Check to see if the Personality is already running

An APP or AM Replacement node requires that its LCN/TPN Personality be loaded before it becomes functional. If all of the following conditions are true, the LCN/TPN personality may already be loaded and functional:

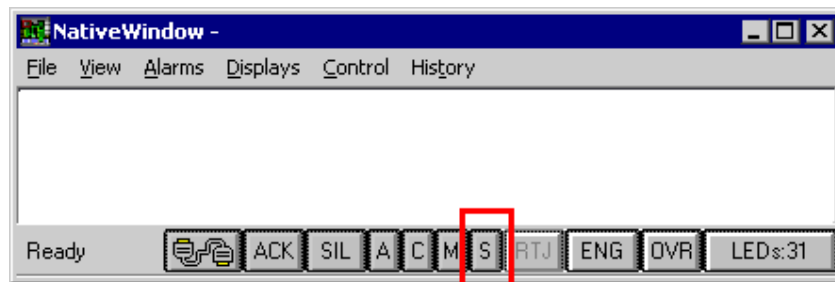
- The node is running with an LCNP4 board.
- The node was not powered down after its last shutdown.

If the System Status Display shows the node to be operational, there is no need to reload the LCN/TPN personality. If the node is not operational, it needs to be reloaded.

To see a typical System Status Display, see Figure 2, "Typical System Status Display."

To load a personality into an APP node

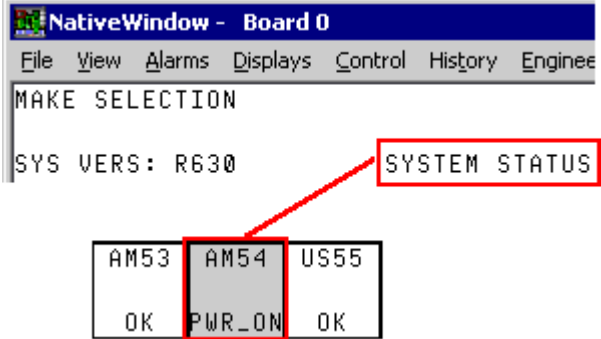
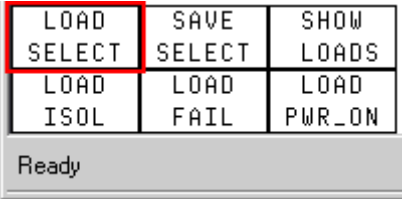
Step	Action
1	Click the S button in the Native Window status bar to open the LCN/TPN System Status Display.



- | | |
|---|---|
| 2 | Select the node to be loaded from the System Status Display. In the following example, node AM54 is selected. |
|---|---|

4. Loading the LCN/TPN Personality

4.2. Loading the Personality for an APP node

Step	Action
	
3	Select the LOAD SELECT target and press the ENTER key. 



REFERENCE - INTERNAL

For details on the LCN/TPN System Status Display, see Section 11 of the *Process Operations Manual*.

Prior to release R650, the APP node could not be loaded without the system HM. With Release R650, you can load the APP or AM node personality from the system HM or a removable media mounted on a GUS/US. The fast-load cartridge can be created on any available physical or emulated media.

GUS nodes can be loaded manually from a fast-load cartridge mounted on a different US/GUS (not the local node) without requiring the system HM to be on-line.

4. Loading the LCN/TPN Personality

4.2. Loading the Personality for an APP node

APP nodes can be reloaded with the APP or AM personality without using a system HM. When the system HM is not available (or when the APP node loses communication with the system HM) and the APP node will be reloaded with the APP personality without system HM, then the personality is loaded from its local disk and the support files are obtained from the fast-load zip disk mounted in the GUS/US. If the APP node is to be reloaded with the AM personality, both the personality and support files are obtained from the fast-load zip disk mounted on GUS/US.



ATTENTION

The fast-load cartridge can contain only one personality—either the APP personality or the AM personality. If both personalities are present in the cartridge, the AM personality is loaded by default. To load the same node with both the personalities, two separate fast-load zip disks are required—one containing the APP personality and another containing the AM personality.

Manual load of an APP node Personality

The following table contains the procedure to load an APP node with the APP Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge - Manual Load.

Procedure Prerequisite: The APP node should be in PWR_ON, or QUALIF, or FAIL state before you can perform a manual load.

Step	Action
1	Mount the APP/AM fast-load cartridge that includes APP personality files (directory &AXM) on a GUS/US. Ensure that the fast-load cartridge does not include AM personality files (directory &AMO).
2	From a US/GUS node on the LCN, press the SYS STATS button to open the System Status display.
3	Select the required APP node.
4	Select NODE STATUS .
5	Select the required APP node.
6	Select LOAD/DUMP .
7	Select MANUAL LOAD .
8	Select required mode from the options HOT LOAD , WARM LOAD , COLD LOAD , and NO POINT PROCESS .
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK , and the node type changes from AM to APP .

4. Loading the LCN/TPN Personality

4.2. Loading the Personality for an APP node

Auto load of an APP node Personality (Local)

The following table contains the procedure to load an APP node with the APP Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge – AutoLoad Local:

Step	Action
1	Mount the APP/AM fast-load cartridge that includes APP personality files (directory &AXM) on a GUS/US. Ensure that the fast-load cartridge does not include AM personality files (directory &AMO).
2	From the US/GUS node on the LCN, press the SYS STATS button to open the System Status display.
3	Select the required APP node.
4	Select NODE STATUS .
5	Select the required APP node.
6	Select LOAD/DUMP .
7	Select AUTOLOAD LOCAL .
8	Press ENTER to execute.
	Note: In some instances where the fast-load cartridge is mounted on the requesting node, it is not necessary to perform steps 9–15.
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK , and the node type changes from AM to APP .

Manual load of an APP node with AM Personality

The following table contains the procedure to load an APP node with the AM Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge - Manual Load:

Step	Action
1	Mount the APP/AM fast-load cartridge that includes AM personality files (directory &AMO) on a GUS/US.
2	From a US/GUS node on the LCN, press the SYS STATS button to open the System Status Display.
3	Select the required APP node.
4	Select NODE STATUS .
5	Select the required APP node.
6	Select LOAD/DUMP target.
7	Select MANUAL LOAD .
8	Select the required mode from the options COLD LOAD , WARM LOAD , HOT LOAD , and NO POINT PROCESS .
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number that contains fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the load is complete, the node status is displayed as OK and the node type is displayed as AM .

4. Loading the LCN/TPN Personality

4.2. Loading the Personality for an APP node

Auto load of an APP node with AM Personality

The following table contains the procedure to load an APP node with the AM Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge – AutoLoad Local:

Step	Action
1	Mount the APP/AM fast-load cartridge that includes AM personality files (directory &AMO) on a GUS/US.
2	From a US/GUS node on the LCN, press the SYS STATS button to open the System Status Display.
3	Select the required APP node.
4	Select NODE STATUS .
5	Select the required APP node.
6	Select LOAD/DUMP target
7	Select AUTOLOAD LOCAL .
8	Press ENTER to execute.
	Note: In some instances where the fast-load cartridge is mounted on the requesting node, it is not necessary to perform steps 9–15.
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK , and the node type is displayed as AM .

5. Shutting Down the LCN/TPN Personality

5.1 Shutdown a Node's LCN/TPN Personality

Use site-specific procedure

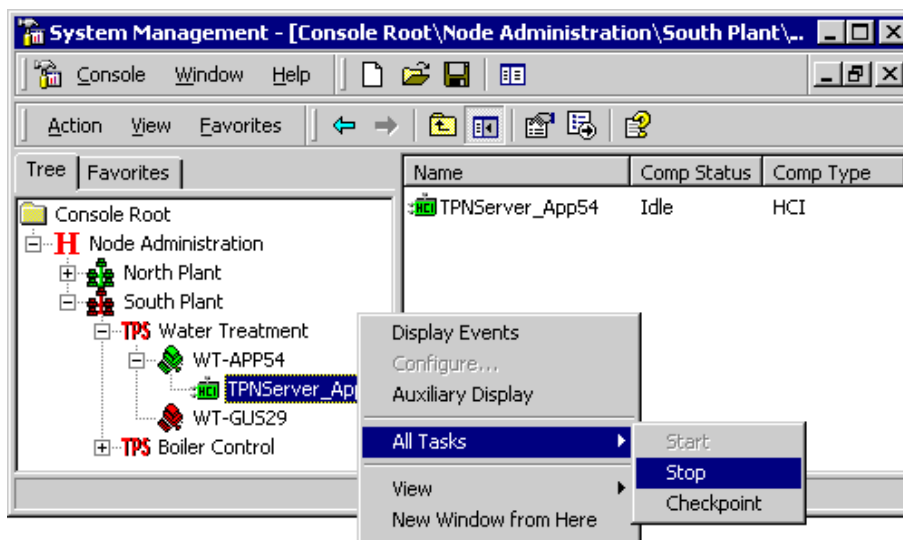


ATTENTION

If your system administrator has provided shutdown procedures, use them to shut down the node. If these have not been provided, then use the following shutdown procedure.

Procedure prerequisites

Nodes may have Honeywell “managed” components or applications that should be safely shut down before shutting down the node’s LCN/TPN Personality. Managed components are shut down from the **System Management** display as shown in the following figure:



REFERENCE - INTERNAL

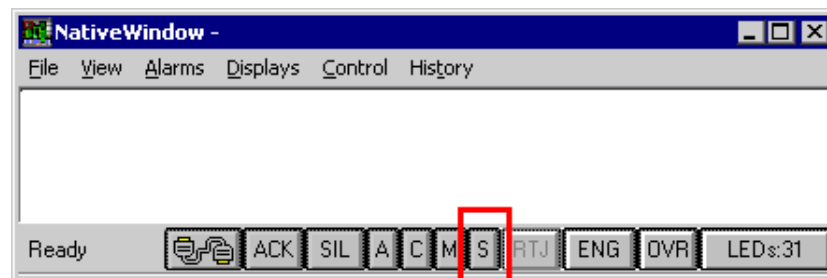
For more information about managed components on the System Management Display, see the "Starting an HCI Managed Component" topic of the *System Management Operations Guide*.

5. Shutting Down the LCN/TPN Personality

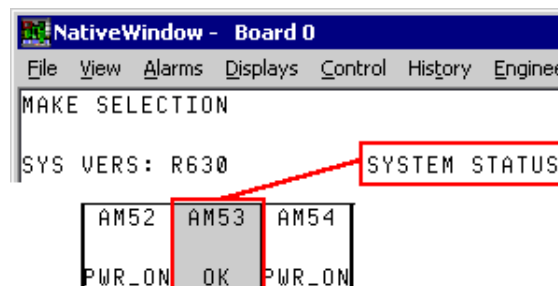
5.1. Shutdown a Node's LCN/TPN Personality

To shut down a node's LCN/TPN Personality

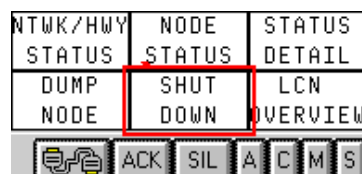
- | Step | Action |
|------|--|
| 1 | Click the S button in the Native Window's tool bar to open the LCN/TPN System Status Display. |



- | | |
|---|--|
| 2 | Select the node to be shut down from the System Status Display. In the following example, node AM53 is selected. |
|---|--|



- | | |
|---|--|
| 3 | Select the SHUT DOWN target and press the ENTER key. |
|---|--|



Shut down Windows only when necessary

After shutting down the LCN/TPN Personality, a Windows shutdown is only necessary to turn off the node. You must shut down Windows at the node to be turned off using standard Window practices (click **Start > Shut Down**) and following your site's operational policies and guidelines.

6. Generating LCN Revision and Version Reports

6.1 Overview of LCN Revision and Version Reports

About LVRLOG

With GUS R360, Experion Station-TPS (ES-T) R300, and TPN R670, LCN Version and Revision reports can be generated using the LCN Node Version/Revision Logging tool (LVRLOG), from a GUS or an ES-T. The tool generates the LVRLOG.RD (text file) and the LVRLOG.RE (csv file), which are automatically stored in History Module directory NET >&CUS.

The LCN Version and Revision report contains a detailed listing about the following LCN devices:

- Each node on the LCN.
- Each Data Hiway connected to the LCN, and each Box on each Data Hiway.
- Each UCN connected to the LCN, each Module on the UCN, and each card in each Module.

The Windows-based application is referred as “LVRLOG application” in this document.



REFERENCE – INTERNAL

For more information about the report, see the *Engineer's Reference Manual*.

Accessing the online help

To access the LVRLOG online help, perform the following steps:

Step	Action
1	Click Start > Run to start the LVRLOG help. Result: The Run dialog box appears.
2	Type LVRLOG.hlp and then click OK . Result: The LVRLOG help file appears.

6. Generating LCN Revision and Version Reports

6.1. Overview of LCN Revision and Version Reports

Installing LVRLOG application

When you install the GUS Display runtime on a

- GUS, then the LVRLOG.EXE file is automatically installed in the Honeywell\TPS\Base folder.
- ES-T, then the LVRLOG.EXE file is automatically installed in the Honeywell\Experion\TPS\Base folder.

Before starting the LVRLOG application

Before starting the LVRLOG application, ensure the UPLVR load module is available on GUS or ES-T.



ATTENTION

- Initiating the LVRLOG is applicable only to GUS and ES-T nodes and is not applicable to APP and Experion Server-TPS (ESVT) nodes.
 - LVRLOG can be started only when the data servers are active.
 - Users with Operator access level can start LVRLOG.
-

To make the most efficient import of the LCN Node Version/Revision text file, Honeywell recommends that you have the File Transfer available on the GUS or ES-T.

6.2 Using LVRLOG application

Syntax of LVRLOG command

The syntax for the LVRLOG application command must be specified through the command-line interface to LVRLOG. The syntax is as follows:

```
LVRLOG [/SITE_ID <SiteId>] [/LCN_ID <LCNId>] [/HEADER] [/ -HEADER]  
[/LCN] [/ -LCN] [/DH] [/ -DH] [/UCN] [/ -UCN]
```

The Site ID and the LCN_ID can have a '/' and a '\' as characters but they must be preceded by a '\'. A '\' must be followed by either a '/' or a '\'. An example of a format, its value, validity, and reason are as follows:

Format	Value	Valid or Invalid	Reason
/Site_ID abV12	ab/12	Valid	
/Site_ID ab\\12	ab\12	Valid	
/Site_ID ab\12		Invalid	A '\' must be succeeded by a '/' or a '\'.
/Site_ID abcd		Invalid	A '\' must be succeeded by a '/' or a '\'.
/Site_ID abcd \\LCN_ID 1234 /HEADER	abcd /LCN_ID 1234	Invalid	A parameter separator '/' is left out because of a '\'. The Site_ID is considered as abcd /LCN_ID 1234.

You can use either LVRLOG or LVRLOG.EXE as a command to start LVRLOG report generation. The following list describes these commands:

- If you use the LVRLOG.EXE command, you can change the report options. If no option is specified, the default values are used in the report generation.
- If you use the LVRLOG command, the default values are used for all the options. For examples on using the LVRLOG command, see the “LVRLOG application usage scenarios” topic.



ATTENTION

You can specify the options in any order, but you cannot specify an option more than once.

6. Generating LCN Revision and Version Reports

6.2. Using LVRLOG application

The following table describes the options of the LVRLOG command:

Option	Description
/SITE_ID	Specifies the Site ID for the report. The Site ID must be alphanumeric characters in the range of 1-8. If the Site ID has a "/" or a "\", it has to be prefixed with a "\" character. For example, instead of "AB1/2", specify as "AB1\\2" and instead of "AB1\\2", specify as "AB1\\\\2". If the /SITE ID option is not specified, the Site ID is blank in the report.
/LCN_ID	Specifies the LCN ID for the report. The LCN ID must be alphanumeric characters in the range of 1-8. If the LCN ID has a "/" or a "\", it must be prefixed with a "\" character. For example, instead of "AB1/2", specify as "AB1\\2" and instead of "AB1\\2", specify as "AB1\\\\2". If the /LCN option is not specified, the LCN ID is blank in the report.
/HEADER	Specifies whether headers are to be included in the report. /HEADER is the default value and specifies that headers are to be included. /-HEADER specifies that headers are not to be included. If neither option is specified, the default is /HEADER.
/LCN	Specifies whether LCN nodes are to be included in the report. /LCN is the default value and specifies that LCN nodes are to be included. /-LCN specifies that LCN nodes are not to be included If neither option is specified, the default is /LCN.
/DH	Specifies whether the Data Hiway boxes are to be included in the report. /DH is the default value and specifies that Data Hiway boxes are to be included. /-DH specifies that DH boxes are not to be included. If neither option is specified, the default is /DH.
/UCN	Specifies whether UCN controllers are to be included in the report. /UCN is the default value and specifies that UCN controllers are to be included.

Initiating LVRLOG application

You can start the LVRLOG application from the Command prompt or a Windows-based application. The application runs in the background mode to generate report files. The Native Window of the node and the other applications continue to run normally.

To start LVRLOG from the Command prompt, perform the following steps:

Step	Action
1	Click Start > All Programs > Accessories > Command Prompt to start the LVRLOG generation. Result: The Command prompt window appears.
2	In the Command prompt window, type LVRLOG.EXE and selected options, or type LVRLOG with no options. For information on LVRLOG options, see the "Syntax of LVRLOG command" topic.
3	Press Enter. Response: The syntax is validated. If the syntax is valid, the Command prompt returns. The LVRLOG application, however, runs in the background to generate report files. The completion status or the error status is available in the Windows Application Log. Though LVRLOG is started from TPS, the report is generated only on the TPN. If the syntax is invalid, an event is logged with the error message. For details on LVRLOG error messages, see Table 4, "LVRLOG Error Description." If the LVRLOG command fails, errors are logged only in the Event Viewer.



REFERENCE – INTERNAL

For details on the initiation and report generation options on LVRLOG, see the *Engineer's Reference Manual*.

6. Generating LCN Revision and Version Reports

6.2. Using LVRLOG application

Exiting LVRLOG application



ATTENTION

The LVRLOG application automatically ends its execution when the report generation is complete. If required, the execution of the LVRLOG can be explicitly ended using Windows Task Manager. In such a case, the LVRLOG ends only on the TPS side. On the TPN side, however, the LVRLOG continues to run until completion.

Using LVRLOG application in user applications

You can start LVRLOG generation from Windows by using LVRLOG.EXE with user applications. The Windows CreateProcess() can be used for creating a process for LVRLOG application.

The following is an example of using the API – CreateProcess() for LVRLOG application:

```
BOOL CreateProcess(  
    "C:\Program Files\Honeywell\TPS\Base\LVRLOG.exe"  
    "/SITE_ID HPS /LCN_ID 123 /HEADER /LCN /UCN /-DH",  
    NULL,                // Process handle not inheritable.  
    NULL,                // Thread handle not inheritable.  
    0,                   // Set handle inheritance to FALSE.  
    NULL,                // No creation flags.  
    NULL,                // Use parent's environment block.  
    NULL,                // Use parent's starting directory.  
    &si,                  // Pointer to STARTUPINFO structure.  
    &pi )                 // Pointer to PROCESS_INFORMATION  
    structure  
);
```

The result of the LVRLOG initiation is returned to the calling program through the Process Exit Code. If the exit code value is zero, it indicates the completion of report generation. Any other exit code value indicates an error. For details on error messages, see Table 4, "LVRLOG Error Description."



ATTENTION

The CreateProcess() starts the process of LVRLOG application and proceeds without waiting for completion.

The initiating application must use Windows process APIs like GetExitCodeProcess() to track the completion of the LVRLOG application. It must also access the exit code to identify whether the LVRLOG generation is successful.

Using LVRLOG application in batch files

LVRLOG application can be used in batch files, which can be scheduled using the Scheduled Tasks feature in Windows.

Example (use the following in a batch file):

```
c:\program files\Honeywell\tps\base\LVRLOG.exe /SITE_ID HPS  
/LCN_ID 123 /HEADER /LCN /UCN /-DH
```



REFERENCE - EXTERNAL

For more details on Scheduled Tasks, see the help on the Microsoft Windows Operating System.

6.3 LVRLOG application events and error messages

LVRLOG application usage scenarios

The following scenarios are applicable to the LVRLOG application:

Table 2 LVRLOG application usage scenarios

Scenario	Explanation
Parent application terminates	After initiating the LVRLOG application, the parent application terminates. However, the LVRLOG application runs until it receives completion status from the TPN side.
LVRLOG application terminates	If the user terminates the LVRLOG application, the TPN side LVRLOG runs to completion. However, the status may not be available to the user.
Examples with different parameters	• LVRLOG – The LVRLOG application is started with the default values for all the parameters. • LVRLOG.EXE /SITE_ID HPS /LCN_ID 1234 – The LVRLOG application is started with the Site ID as HPS, the LCN ID as 1234, and default values for all the other parameters. • LVRLOG.EXE /UCN /LCN /-DH –The LVRLOG application report includes UCN and LCN nodes, but excludes DH boxes. The Site ID and LCN ID are blank.

LVRLOG application events

Events are logged during the “Start” and “End” of the LVRLOG in the Application log of the Windows Event Viewer. The description of the events logged while terminating the LVRLOG are included in the status/error number. For details on error messages, see Table 4, “LVRLOG Error Description.”



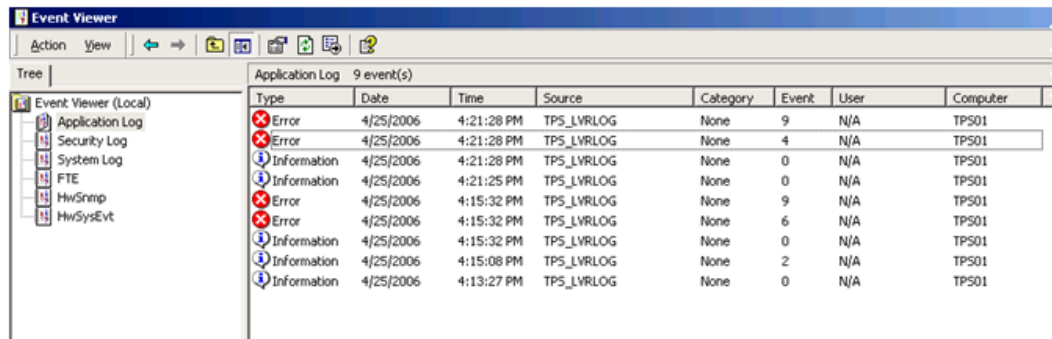
ATTENTION

Events are not logged if you end LVRLOG application manually from the Windows Task Manager.

6. Generating LCN Revision and Version Reports

6.3. LVRLOG application events and error messages

The following figure illustrates an example of how LVRLOG application events are logged in the Windows Event Viewer. To see the event description, right-click the event and then select **Properties**.



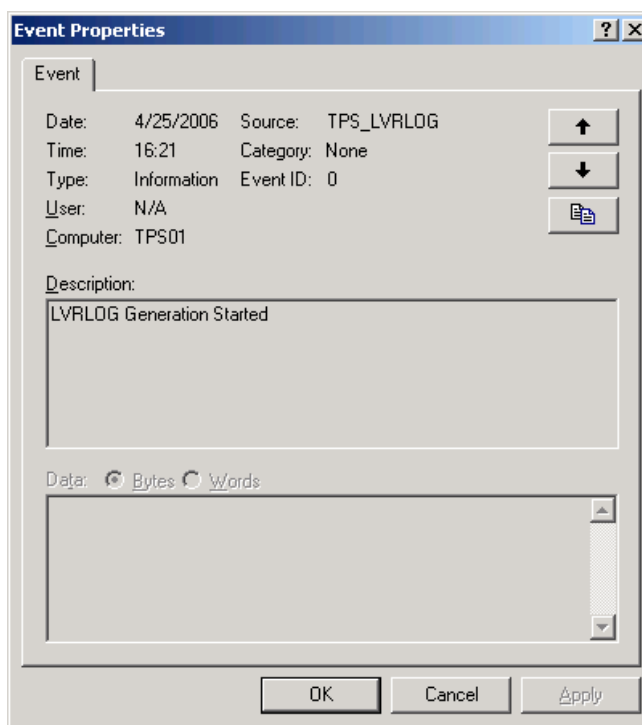
6. Generating LCN Revision and Version Reports

6.3. LVRLOG application events and error messages

The following table lists and describes the properties of LVRLOG application events:

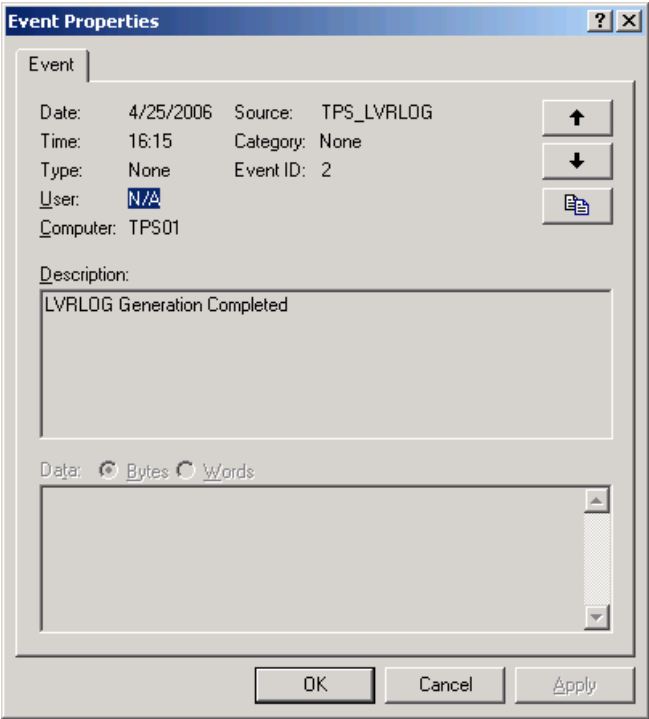
Table 3 Event Description

Event Description	Significance
LVRLOG generation started.	LVRLOG application is started. The following figure illustrates the properties of this event.



6. Generating LCN Revision and Version Reports

6.3. LVRLOG application events and error messages

Event Description	Significance
LVRLOG generation completed.	LVRLOG report generation was completed successfully and the application has stopped execution. 

6. Generating LCN Revision and Version Reports

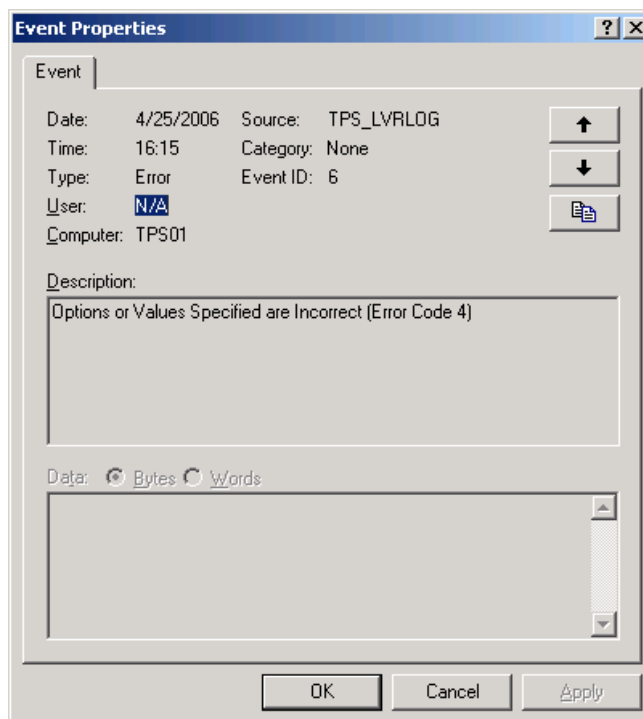
6.3. LVRLOG application events and error messages

Event Description

Options or values specified are incorrect. [Error code<Error Number>]

Significance

The syntax of the LVRLOG command is incorrect. The following figure illustrates the properties of this event.



For more information about the error, see Table 4, "LVRLOG Error Description."

When the LVRLOG command fails, specific events are logged and the LVRLOG application stops executing.

6. Generating LCN Revision and Version Reports

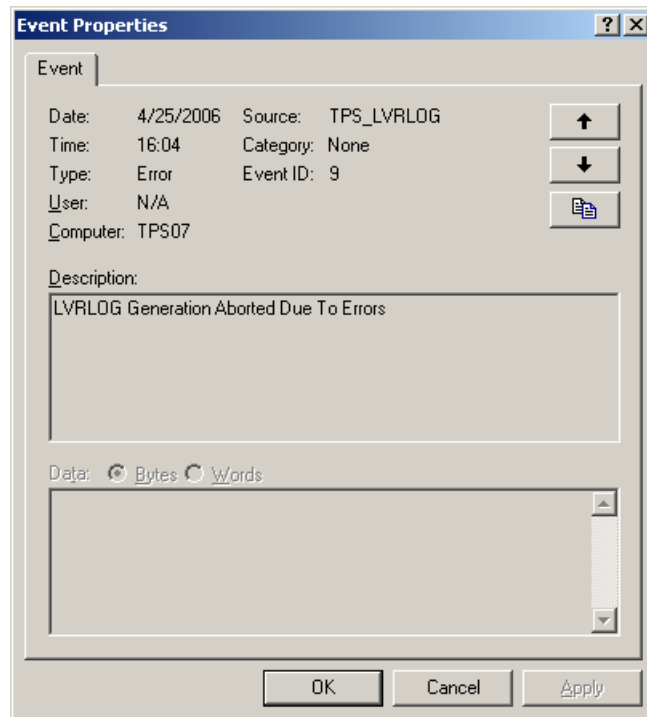
6.3. LVRLOG application events and error messages

Event Description

LVRLOG generation aborted due to errors.

Significance

The LVRLOG application has stopped executing and no report was generated. The following figure illustrates the properties of this event.



6. Generating LCN Revision and Version Reports

6.3. LVRLOG application events and error messages

LVRLOG application error messages

If the LVRLOG command fails, you may encounter the following error messages:

Table 4 LVRLOG Error Description

Error Number	Error Description	Remedial Action
001	Node types are not selected.	Select at least one Node type.
002	Load module or GUS personality not loaded or data server is offline.	Load UxS with the UPLVR load module. Ensure the data server is not online.
003	On more than one initiation the following happens: • LVRLOG application starts from TPS when TPN schematics-based LVRLOG is in progress. • LVRLOG application starts from TPS when another instance of LVRLOG.EXE is running • LVRLOG application starts and then ends (using task manager or using APIs) while the generation is still in progress and another instance of LVRLOG application starts while the TPN side LVRLOG is still running.	Previous LVRLOG request is still running. Wait until the request is completed.
004	Options or values specified are incorrect. Example: LVRLOG.EXE /SITEID xyz /LCN_ID 123	Provide proper arguments.
005	Conflicting options or same option is specified more than once. Example: LVRLOG.EXE /UCN /-UCN	Provide proper arguments.

6. Generating LCN Revision and Version Reports
6.3. LVRLOG application events and error messages

Error Number	Error Description	Remedial Action
006	Site ID or LCN ID value exceeds the maximum number of characters.	Re-type the command with Site ID or LCN ID not exceeding eight characters.
007	LVRLOG files are protected or open.	Unprotect or close LVRLOG.RD and LVRLOG.RE files.
008	Load module or GUS personality is not loaded or data server is offline or previous instance of LVRLOG is running.	Load UPLVR module or wait until the previous instance of LVRLOG completes its execution. Ensure data server is online.
009	LVRLOG generation aborted due to errors.	Provide proper arguments and try again.
100	Internal errors.	N/A

6. Generating LCN Revision and Version Reports

6.3. LVRLOG application events and error messages

7. Using the GUS Logic tool

7.1 Overview

About the GUS Logic tool

With GUS R360 and Experion PKS R310 Experion Station-TPS (ES-T) nodes, the xPM Logic display is introduced for viewing graphical representations of xPM logic points. The graphical representation consists of the inputs, logic blocks, and outputs associated with the point.



REFERENCE - INTERNAL

For information about scripting the xPM Logic display, see the *GUS Display Scripting User's Guide*.

Configuring the xPM Logic display

Based on the screen resolution of the GUS node, two versions of the xPM Logic display are provided. To configure the xPM Logic display, perform the following steps:

Step	Action
1	Copy the xPM Logic display file (.pct) from the standard display folder to the necessary folder on the GUS node. For example, copy the xpmlogic_display.pct file to c:\displays.
	ATTENTION If the screen resolution of your GUS node is 1280x1024 or higher, use the xpmlogic_display.pct. If the screen resolution of your GUS node is 1024x768, use the xpmlogic_display_lores.pct.
2	Rename the display file such that the file name does not exceed eight characters.
3	Configure the display folder in the GUS Displays search path.
	REFERENCE - INTERNAL To configure the display search path, see the <i>System Components Configuration Utility User's Guide</i> .

Prerequisites for using the xPM Logic display

Before using the Logic tool, ensure the following:

- The xPM point is already configured and loaded.
- LCN data server status is OK.

7.2 Using the xPM Logic display



ATTENTION

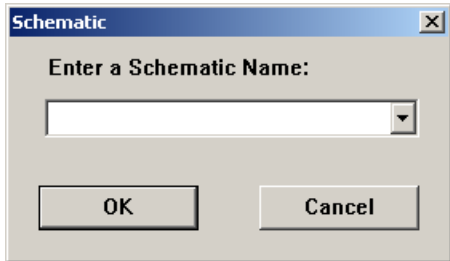
- The xPM Logic display is used only for viewing a graphical representation of the xPM logic point.
 - Process values are dynamically updated in the display.
 - If a point is reconfigured, you must refresh the display to view the updated values.
 - The input/output labels are blank if the xPM logic point is configured with hardware addressing.
-

Calling up the xPM Logic display

The xPM Logic display can be called in the following ways:

- Using the SCHEM button
- Using InvokeDisplay
- Using targets that are scripted with a pre-selected tag

To call up the xPM Logic display using the SCHEM button

Step	Action
1	On the GUS/ES-T node, press SCHEM . The Schematic dialog box appears. 
2	Type the display name and then click OK . The xPM Logic display window appears.
3	In the Tag Name , type the tag name of the associated xPM logic point and press Enter . If the tag name is invalid, an error message appears.

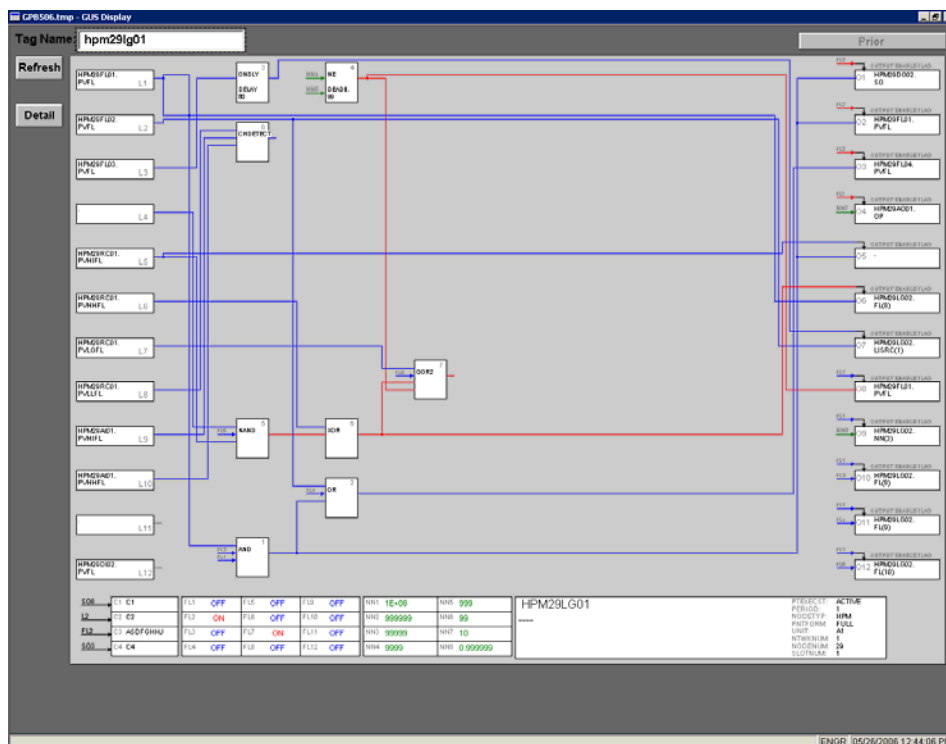
7. Using the GUS Logic tool

7.2. Using the xPM Logic display

Step

Action

- If the tag name is valid, a graphical representation of the logic point appears. The process values are dynamically updated.



ATTENTION

Alternately, you can also call up the display using InvokeDisplay and then specify the tag name to be viewed.

The display can also be called up through targets that are scripted with a pre-selected tag.



REFERENCE - INTERNAL

For information about scripting the targets with a pre-selected tag, see the *GUS Display Scripting User's Guide*.

To call up the xPM logic display using targets scripted with a pre-selected tag, perform the following steps:

Step	Action
1	On the GUS node, call up the GUS display that has a target scripted to call up the xPM Logic display.
2	Click the target to call up the graphical representation of the xPM Logic point.

xPM Logic display operations

The following operations can be done using the xPM Logic display:

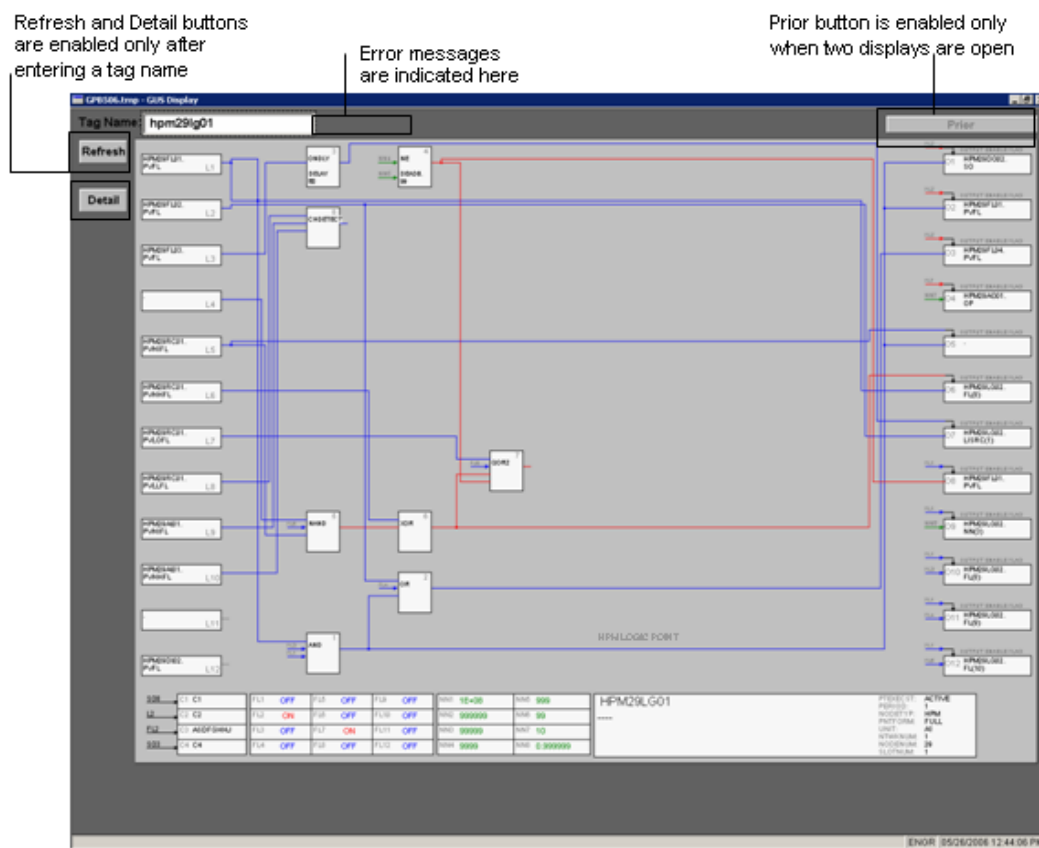
xPM Logic display operation	Action
Call up the Native Window detail display of the xPM logic point.	Click Detail .
Call up the detail display of the input/output points.	Click the input/output labels.
Go back to an xPM Logic display that was previously opened.	Click Prior .
	 ATTENTION The Prior button is enabled only when two xPM Logic displays are open. In addition, the tag name from the previously opened display is also shown on the button.
Update the display.	Click Refresh . Note that configuration values are not updated dynamically.

7. Using the GUS Logic tool

7.2. Using the xPM Logic display

The following figure shows the xPM Logic display interface:

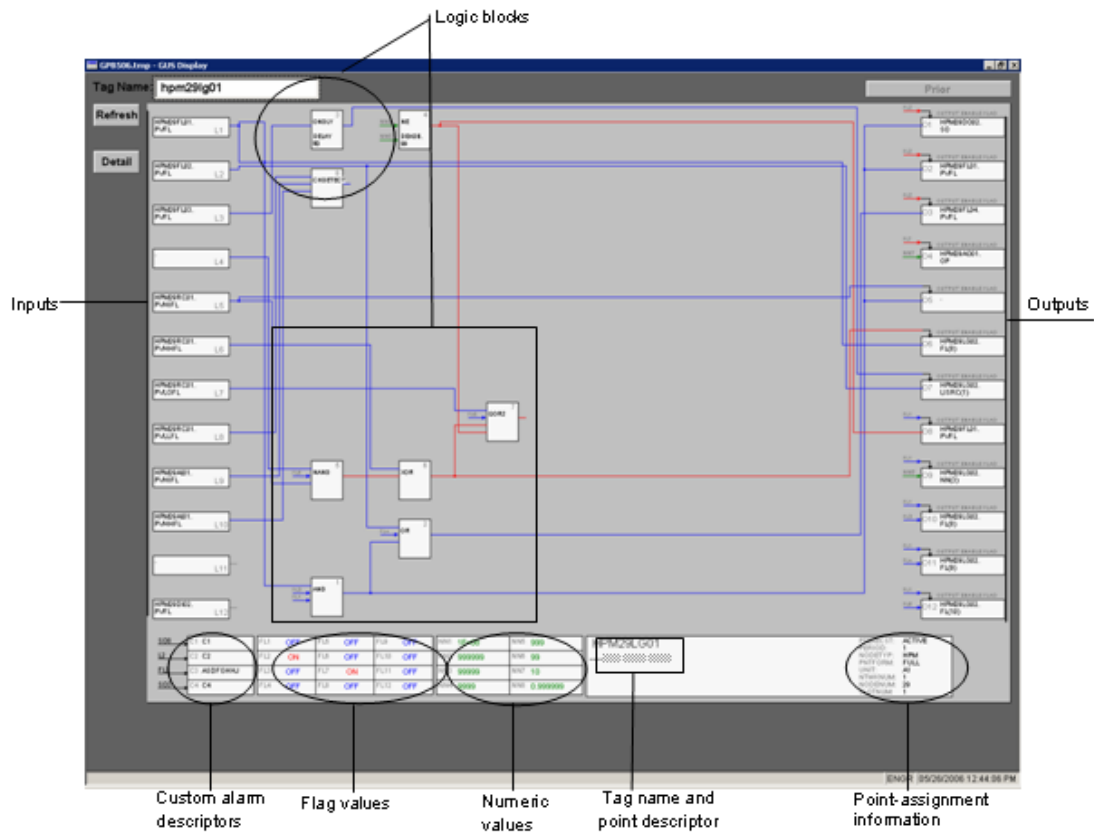
Figure 4 xPM Logic display interface



Elements of an xPM Logic display

The following figure shows the graphical representation of an xPM logic point:

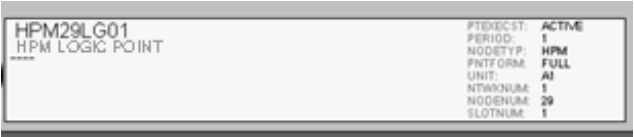
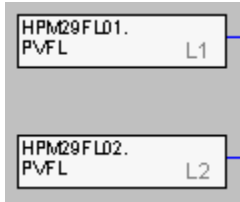
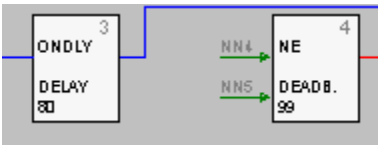
Figure 5 Elements of an xPM Logic display

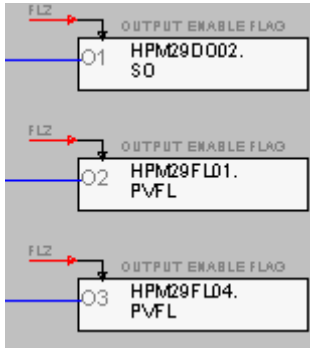
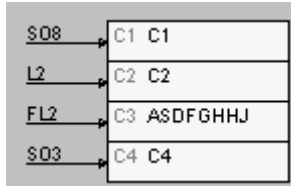


7. Using the GUS Logic tool

7.2. Using the xPM Logic display

The graphical representation of an xPM logic point consists of the following items:

Items	Description
Point assignment information	Includes tag name, point descriptor, PTEXCST, PERIOD, NODETYP, PNTFORM, UNIT, NTKWNUM, NODENUM, and SLOTNUM parameters. 
Inputs	Inputs are labeled as L1, L2, L3, and so on. The sources (tag names) of the inputs are also indicated. 
Logic blocks	The logic blocks are numbered as 1, 2, 3, and so on. The following information is shown for the logic blocks: - Name of the logic block - Deadband value for EQ, NE, GT, LT, and LE blocks - Delay time value for PULSE, MINPULSE, MAXPULSE, ONDLY, OFFDLY, WATCHDOG, and DISCREP3 blocks  The number of inputs, outputs, and logic blocks are derived from the Logic Mix (LOGMIX) selection during point configuration.

Items	Description
Outputs	Outputs are labeled as O1, O2, O3, and so on.  The following information is shown for the outputs: • Output enable flag • Tag name and parameter of Destination point
Connections	Logic connections are shown as blue and red lines. Red lines represent ON values and blue lines represent OFF values. Analog connections are shown as green lines.
Custom alarm descriptors	Custom alarm descriptors are indicated as C1, C2, C3, and so on. The following information is indicated for custom alarm descriptors: • Descriptor • Source 

7. Using the GUS Logic tool

7.2. Using the xPM Logic display

Items	Description																								
Flag values	Indicates current flag values. <table><tr><td>FL1</td><td>OFF</td><td>FL5</td><td>OFF</td><td>FL9</td><td>OFF</td></tr><tr><td>FL2</td><td>ON</td><td>FL6</td><td>OFF</td><td>FL10</td><td>OFF</td></tr><tr><td>FL3</td><td>OFF</td><td>FL7</td><td>ON</td><td>FL11</td><td>OFF</td></tr><tr><td>FL4</td><td>OFF</td><td>FL8</td><td>OFF</td><td>FL12</td><td>OFF</td></tr></table>	FL1	OFF	FL5	OFF	FL9	OFF	FL2	ON	FL6	OFF	FL10	OFF	FL3	OFF	FL7	ON	FL11	OFF	FL4	OFF	FL8	OFF	FL12	OFF
FL1	OFF	FL5	OFF	FL9	OFF																				
FL2	ON	FL6	OFF	FL10	OFF																				
FL3	OFF	FL7	ON	FL11	OFF																				
FL4	OFF	FL8	OFF	FL12	OFF																				
Numeric values	Indicates current numeric values. <table><tr><td>NN1</td><td>1E+08</td><td>NN5</td><td>999</td></tr><tr><td>NN2</td><td>999999</td><td>NN6</td><td>99</td></tr><tr><td>NN3</td><td>99999</td><td>NN7</td><td>10</td></tr><tr><td>NN4</td><td>9999</td><td>NN8</td><td>0.999999</td></tr></table>	NN1	1E+08	NN5	999	NN2	999999	NN6	99	NN3	99999	NN7	10	NN4	9999	NN8	0.999999								
NN1	1E+08	NN5	999																						
NN2	999999	NN6	99																						
NN3	99999	NN7	10																						
NN4	9999	NN8	0.999999																						



REFERENCE - INTERNAL

For more information about the description of xPM logic point elements, see the following documents:

- *Process Manager Control Functions and Algorithms—pm09*
 - *Advanced Process Manager Control Functions and Algorithms—ap09*
 - *HPM Control Theory—hp09*
-

7.3 Troubleshooting the xPM Logic display

Error messages

Before using the xPM Logic display it is important to ensure that all prerequisites for using the tool are addressed. Error messages associated with the xPM Logic display appear in red background and are located next to the tag name. Communication errors are also indicated by the PTEXECST parameter on the display area.

Display area does not indicate any values

Description	"@@@@@" values are indicated for inputs and outputs. PTEXECST indicates "retrieving."
Cause	You are trying to call up the display for the first time. OR The xPM node is in the process of reloading.
Resolution	If you are calling up the display for the first time, wait for a few seconds. Ensure the xPM node is loaded. The display is automatically updated after the node is reloaded.

Communication error with LCN

Description	PTEXECST indicates "Communication error."
Cause	The associated xPM is in a failed state or the LCN data server is down.
Resolution	Ensure that the xPM node is running and the status of the LCN data server is OK.

7. Using the GUS Logic tool

7.3. Troubleshooting the xPM Logic display

Tag not found

Description	"Tag Not Found" error message appears.
Cause	The tag name is invalid. OR The associated xPM or NIM is not running.
Resolution	Before calling up the xPM Logic display, ensure the following: • The point is already configured and loaded. • You are using a valid tag name. • The associated xPM or NIM is running.

8. Viewing and Analyzing LCN Load

8.1 LCNLOAD Display

Overview

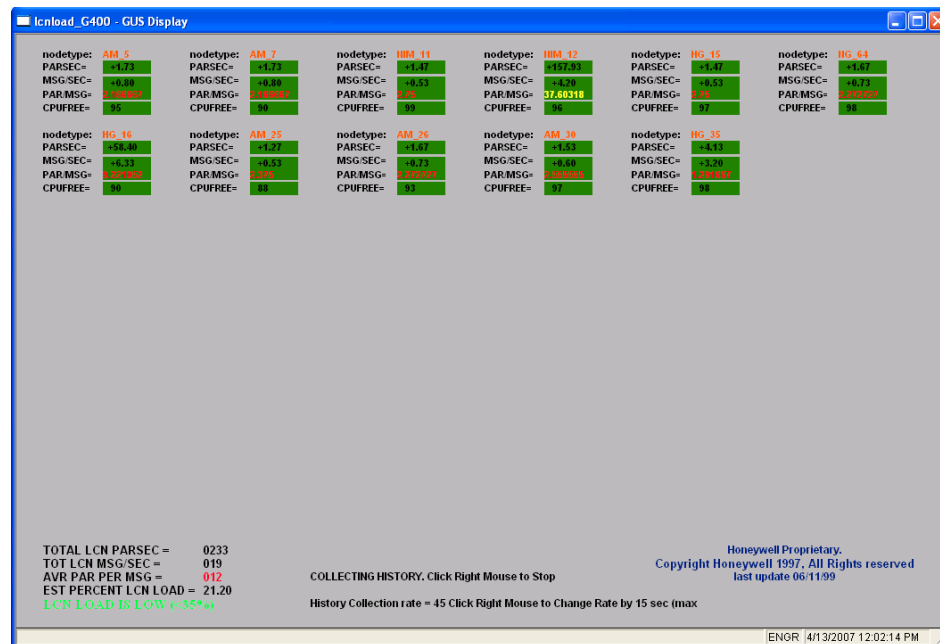
With TPS 340, LCNLOAD_G400.PCT is provided as a Reusable Application Component (RAC) display utility. This display is used for viewing/analyzing the total load on the LCN and on the individual nodes (AM/NIM/HG/CG/NG). The .PCT file is available at C:\Program Files\Honeywell\TPS\RAC\Utility Displays.

The display collects historical load and performance data into .CSV files. The data helps in monitoring long-term network performance.



ATTENTION

The suffix G400 in the filename LCNLOAD_G400.PCT indicates the release number and it changes for every release.



8. Viewing and Analyzing LCN Load

8.1. LCNLOAD Display

Calculating LCN Load

With R680, the LCN load is calculated using a formula and the calculated value is displayed through the LCNPERF display. The display can be accessed through the target LCNPERF in PERFMENU.

Load Calculation Formula

The LCN load calculation tool runs on the US node. The PSDP parameter \$CALCPRD enables the load calculation task and controls the periodicity of load calculations.



ATTENTION

Honeywell recommends enabling the load calculation task only on one or two nodes per segment, as it accesses more than 200 parameters.

The following formula is used for calculating the total estimated percentage of load on LCN.

Formula 1: For calculating the total Estimated % of Load on LCN

Percentage load on LCN = $\frac{(\text{total_par_sec} + 1500 + (\text{msg_per_sec} * 7))}{8800} * 100 * (\$PARLIMX(0)/4500) + \$PARLIMC(0)$.

where,

total_par_sec is the total number of parameters (on AM, NIM, HG, CG and NG nodes) per second.

msg_per_sec is the number of messages (on AM, NIM, HG, CG and NG nodes) per second.

Note: The Base load is calculated as follows:

$(1500/8800)*100 = 17.05 \%$

The base load is 17.05% for a typical traffic on LCN for non-data access. When multiple segments are configured on a network, the base load of 17.05 will be distributed amongst the segments. This distribution helps to avoid the base load being reckoned repeatedly.



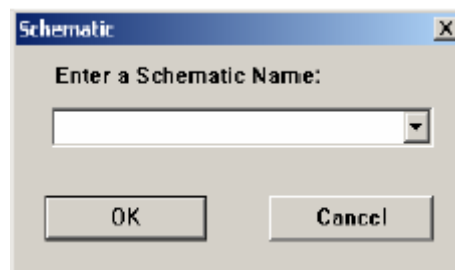
REFERENCE - INTERNAL

For details about the LCNPERF display, see Section 23.6 of the *Engineer's Reference Manual*, sw09-605.

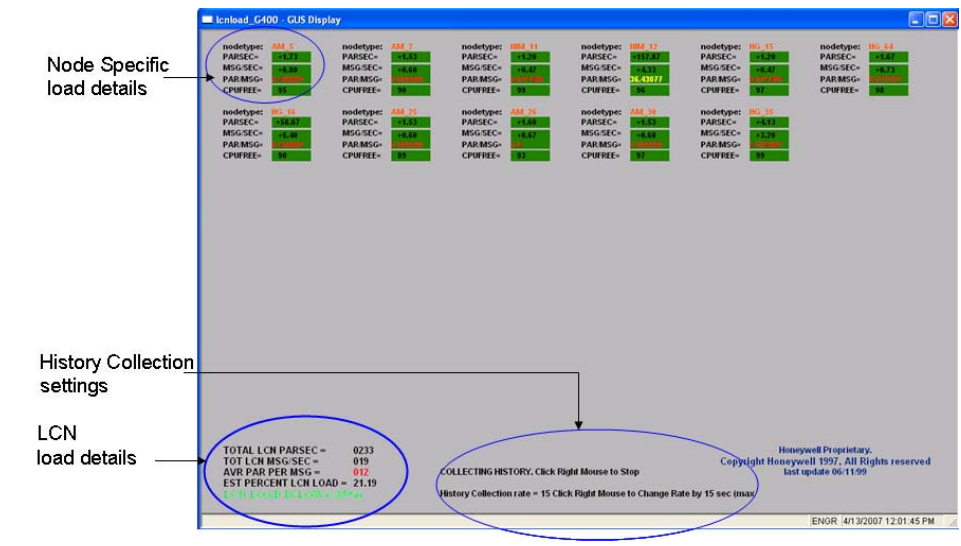
Invoking LCNLOAD display

To invoke the LCNLOAD display, perform the following steps:

Step	Action
1	Press SCHEM key on the keyboard. The Schematic dialog box appears.

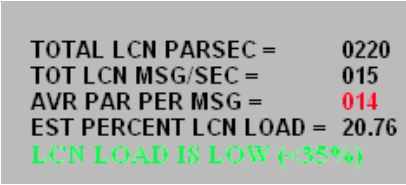
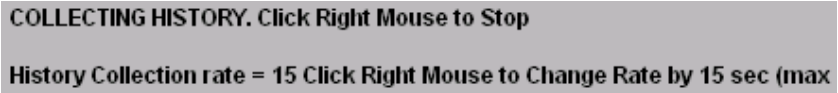


- 2 Type the display name LCNLOAD_G400 and click **OK**.
- The **LCNLOAD** display window appears.



8. Viewing and Analyzing LCN Load

8.1. LCNLOAD Display

Step	Action
Node Specific load details	
LCN load details	
History Collection settings	

LCNLOAD display Operation

The LCNLOAD_G400.PCT display compares individual load parameters with trip limits as specified in the PSDP parameters (\$PPMLOTP, \$PPMLLTP, \$LDHITP, \$LDHHTP) of that node, and then displays the load details.

The following sequence of operations is performed when the LCNLOAD_G400.PCT display is invoked:

- The display scans all the 96 LCN node addresses to check whether the nodes such as AM, NIM, HG, CG and NG exist.
- The display scans the identified nodes for every 4 seconds to determine the status of the nodes based on the performance.
- The load parameters outside the trip limits are indicated in different colors as Warning or Error.



REFERENCE - INTERNAL

For more details on Warnings and Error indications, see [Warnings and Error indications](#).

- The LCNLOAD display shows the LCN load details and the load details of the individual nodes.

Node load details

The LCNLOAD display shows the following values for each node:

Value	Description
PARSEC	Total parameters per second delivered to the LCN
MSG/SEC	Total messages per second delivered to the LCN
PAR/MSG	Average number of parameters per message
CPU FREE	CPUFREE for the node

LCN load details

The LCNLOAD display shows the following values for the entire LCN:

Value	Description
TOTAL LCN PARSEC	Total parameters per second passed through the LCN
TOT LCN MSG/SEC	Total messages per second
AVR PAR PER MSG	Average parameters per message
EST PERCENT LCN LOAD	Estimated load % of LCN (load status of the LCN)

8.2 Warnings and Error indications

Overview

The LCNLOAD display shows the following indications:

- Warning indications are displayed when the load parameter values are less than the specified limits.
- Error indications are displayed when the load parameter values exceed the specified limits.

These indications are used for analyzing the performance problems and improve the performance of the LCN nodes.

Node-specific warning and error indications

Warning (yellow)

The data values in yellow indicate the following:

- Heavy load in the nodes
- Inefficient use of the TPS network nodes



The screenshot shows a node status display for 'IHM_12'. The values for PARSEC, MSG/SEC, and PAR/MSG are highlighted in yellow, indicating a warning. CPUFREE is 96.

nodetype:	IHM_12
PARSEC=	+157.87
MSG/SEC=	+4.33
PAR/MSG=	36.43077
CPUFREE=	96

Warning indication

Error (red)

The data values in red indicate the following:

- Node OVER LOAD – Node reaches its capacity limits and causes network retries.
- Excessive msg rate – Node receives individual messages at a high rate.
- Low msg efficiency – Node receives messages that are packed inefficiently.



The screenshot shows a node status display for 'AM_26'. The values for PARSEC, MSG/SEC, and PAR/MSG are highlighted in red, indicating an error. CPUFREE is 93.

nodetype:	AM_26
PARSEC=	+1.60
MSG/SEC=	+0.67
PAR/MSG=	2.4
CPUFREE=	93

Error indication

LCN warning and error indications

Warning (yellow)

The data values in yellow indicate the following:

- Heavy load in the LCN
- Inefficient use of the TPS network bandwidth

```
TOTAL LCN PARSEC = 0227
TOT LCN MSG/SEC = 017
AVR PAR PER MSG = 013
EST PERCENT LCN LOAD = 20.99
LCN LOAD IS LOW (<35%)
```

Warning indication

Error (red)

The data values in red indicate the following:

- LCN bandwidth is very high
- LCN network message efficiency is extremely low

```
TOTAL LCN PARSEC = 0233
TOT LCN MSG/SEC = 019
AVR PAR PER MSG = 012
EST PERCENT LCN LOAD = 21.19
LCN LOAD IS LOW (<35%)
```

Error indication

List of messages and default limits

The following table lists the default limits of the LCNLOAD display, and the color and status indications that appear in the display if a parameter value exceeds its limit.

Parameter	Limits	Color indication	Node Status
CPUFREE	<=40	Yellow	Node : Heavy Load
	<=20	Red	Node : Over Load
MSG PER SEC	> 35	Yellow	High msg rate
	>40	Red	Excessive msg rate
parpermsg	<100	Yellow	Low msg efficiency
	<20	Red	Low msg efficiency
	>100	Green	Moderate msg efficiency
LCNLoad	>=55	Yellow	LCN load is heavy

8. Viewing and Analyzing LCN Load

8.3. Collecting Historical data for LCN Load

Parameter	Limits	Color indication	Node Status
	>=65	Red	LCN load is very heavy
	<55	Green	Load is moderate
	<35	Green	LCN load is low

8.3 Collecting Historical data for LCN Load

Overview

The LCNLOAD display collects historical data for the load and performance of the nodes on LCN. The data helps in monitoring long term network performance.

The following run time options in the display are used for collecting historical data:

- Setting history collection rate
- Enabling/Disabling history collection

The collected historical data is stored in files *.csv.

Setting history collection rate

To change the rate of history collection, perform the following steps:

Step	Action
1	Right click History Collection rate. • The collection rate increases by 15 seconds for every right click. COLLECTING HISTORY. Click Right Mouse to Stop History Collection rate = 45 Click Right Mouse to Change Rate by 15 sec (max  ATTENTION Ensure the history collection rate is set, before collecting the historical data.

Enabling/Disabling history collection

To enable/disable the history collection, perform the following steps:

Step	Action
1	Right click History Collection rate to enable the history collection. The display starts collecting historical data for every 15 seconds. COLLECTING HISTORY. Click Right Mouse to Stop History Collection rate = 15 Click Right Mouse to Change Rate by 15 sec (max
2	Right click History Collection rate to disable the history collection. COLLECTING HISTORY. Click Right Mouse to Stop History Collection rate = 45 Click Right Mouse to Change Rate by 15 sec (max

8. Viewing and Analyzing LCN Load

8.3. Collecting Historical data for LCN Load

Accessing historical LCN Load details

The collected historical data is stored in *.csv files as shown in the following figure:

	A	B	C	D	E	F	G	H	I	J
1	15	231	21.14	13	date/time = parsec	%load	par/msg)			
2	30	236	21.33	12	date/time = parsec	%load	par/msg)			
3	45	233	21.2	12	date/time = parsec	%load	par/msg)			
4	60	232	21.2	12	date/time = parsec	%load	par/msg)			
5	75	233	21.21	12	date/time = parsec	%load	par/msg)			
6	90	233	21.2	12	date/time = parsec	%load	par/msg)			
7	105	233	21.2	12	date/time = parsec	%load	par/msg)			
8	120	234	21.23	12	date/time = parsec	%load	par/msg)			
9	135	234	21.27	12	date/time = parsec	%load	par/msg)			
10	150	233	21.19	12	date/time = parsec	%load	par/msg)			
11	165	235	21.3	12	date/time = parsec	%load	par/msg)			
12	180	233	21.2	12	date/time = parsec	%load	par/msg)			
13	195	233	21.2	12	date/time = parsec	%load	par/msg)			
14	210	233	21.2	12	date/time = parsec	%load	par/msg)			
15	225	235	21.29	12	date/time = parsec	%load	par/msg)			
16	240	233	21.2	12	date/time = parsec	%load	par/msg)			
17	255	234	21.26	12	date/time = parsec	%load	par/msg)			
18	270	234	21.23	12	date/time = parsec	%load	par/msg)			
19	285	233	21.28	12	date/time = parsec	%load	par/msg)			
20	300	233	21.28	12	date/time = parsec	%load	par/msg)			
21	315	234	21.27	12	date/time = parsec	%load	par/msg)			
22	330	233	21.2	12	date/time = parsec	%load	par/msg)			
23	345	233	21.2	12	date/time = parsec	%load	par/msg)			
24	360	233	21.2	12	date/time = parsec	%load	par/msg)			
25	375	235	21.29	12	date/time = parsec	%load	par/msg)			
26	390	233	21.2	12	date/time = parsec	%load	par/msg)			
27	405	233	21.27	12	date/time = parsec	%load	par/msg)			
28	420	234	21.23	12	date/time = parsec	%load	par/msg)			
29	435	234	21.27	12	date/time = parsec	%load	par/msg)			
30	450	234	21.22	12	date/time = parsec	%load	par/msg)			
31	465	233	21.28	12	date/time = parsec	%load	par/msg)			
32	510	234	21.27	12	date/time = parsec	%load	par/msg)			
33	525	233	21.19	12	date/time = parsec	%load	par/msg)			
34	540	235	21.3	12	date/time = parsec	%load	par/msg)			

Note: The lcnload.csv file is stored at C:\Program Files\Honeywell\TPS\RAC\Templates.

Honeywell

Honeywell International
Process Solutions
2500 West Union Hills
Phoenix, AZ 85027